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RESEARCH ARTICLE

Interpretable Machine Learning Models for PISA Results in Mathematics

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ABSTRACT The Program for International Student Assessment (PISA) 2022 provides a global framework for assessing educational performance worldwide. We addressed a principled observation of characteristics and disparities in Mathematics performance among Spanish adolescents while showing the factors influencing these results. To address this question, we proposed using advanced Machine Learning techniques through possibly non-linear predictive models that identify key drivers of Mathematics performance to inform data-driven educational policies and interventions that improve learning outcomes. By preprocessing the PISA dataset, we categorized students into Low, Medium, and High proficiency levels and employed various binary classification models to discern predictive patterns. In addition, a stacking meta-model integrating the strengths of eight distinct predictive models was developed to enhance prediction accuracy. Our results demonstrated that the meta-model outperforms individual models in predicting student performance across various proficiency levels, consistently showing superior metrics in Precision, Recall, and Area Under Curve (AUC) scores. Specifically, the meta-model achieved an AUC score of 0.9766 when classifying students in the Low and High proficiency categories. We adopted the Shapley Additive exPlanations method to demystify the model decisions, highlighting significant predictors such as grade repetition, access to digital devices, and extracurricular Mathematics classes. We also introduced an interactive dashboard, harnessing Uniform Manifold Approximation and Projection for dimensionality reduction and enabling a granular view of the educational landscape. The intention of all this is to contribute to an education system that is well-informed and effectively adapts to the diverse needs of students.

INDEX TERMS Programme for international student assessment, education, interpretable machine learning, Shapley additive explanations, meta-modeling approach, dashboard visualization.

I. INTRODUCTION

One of the most pivotal evaluations of adolescent education globally is the Programme for International Student Assessment (PISA) 2022 [1], [2]. This triennial international assessment measures 15-year-old students' acquisition of crucial knowledge and skills necessary for successful societal integration. The 2022 iteration of PISA extends its focus to Reading, Mathematics, and Science, incorporating innovative domains alongside the assessment of student well-being. These insights are instrumental in benchmarking student performance across the globe and evaluating the relative efficacy of educational systems among participating nations. The findings, especially in Mathematics and Science, are critical, underscoring their foundational role in the technological advancement of a country [3], [4].

In the contemporary research milieu, an undeniable demand exists for in-depth exploration and sophisticated analysis. Several studies have leveraged on PISA metrics to prove specific dimensions, and a burgeoning segment is exploring the expansive capabilities of Machine Learning (ML) [5], [6], [7], [8], [9], [10], [11]. However, a significant gap persists, as the academic sphere needs a non-linear data-driven models approach capable of navigating the extensive

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data landscape to extract and highlight critical variables [12], [13], [14]. Such a methodology is crucial as we navigate the threshold of a data-centric epoch to offer profound insights while being accessible to a broad spectrum of stakeholders, including researchers and educators [15].

In this paper, PISA 2022 data, focused on Spanish students, is elaborated to define and study three types of students in line with the structure used in the official PISA reports classification system: Low, Medium, and High. Low proficient students show mere comprehension of fundamental mathematical ideas and simple arithmetics. Students at the Medium level understand underlying mathematical concepts to the extent that they can use them for problem-solving. High achievers in Mathematics have the most profound understanding of the subject, with apt complex reasoning, abstractive thinking, and application of sophisticated geometry concepts. In order to develop a simple but also robust analytical framework to classify the students' performance, we decided to adopt three classes (Low, Medium, and High). Similar categorization methods have also been used in earlier studies [16], [17], which have proved to assist in drawing attention to important trends and improving comprehension. This does help in preserving congruence with other processes and patterns of educational research while allowing for greater scrutiny of educational phenomena.

Our main goal is to address the worldwide differences in Mathematics achievement among Spanish adolescents using the latest 2022 PISA data. We apply predictive ML models for attribution analysis of Mathematics performance, mainly concentrating on reading accessibility, critical thinking skills, and geographical aspects through interpretability methods. We explored different binary classification models using six hyperparameter combinations in eight models to find the most effective model for understanding the variables in our study. All models were tuned separately using grid search in every cross-validation fold to find the best set of hyperparameters for the model and enhance performance, which was done in a customized manner. Then, the top eight models that performed well with their optimal hyperparameters in the preceding stage were embedded into a meta-model using the stacking technique. Methods have been developed that utilize the meta-model to combine the predictions of each of the individual models so that overall accuracy is improved and the individual models are quite robust, thus building a strong ensemble model [18], [19]. Next, we explored which factors influence mathematical performance using advanced analytical techniques as Shapley Additive exPlanations (SHAP) for model analysis [20]. This ML approach helps us better understand how these factors affect students' skills. Interpretability models revealed the various factors that impact students' mathematics scores, including access to reading materials, critical thinking skills, gender differences, and location.

The contributions of this work are as follows:

- Eight machine learning models were applied and optimized to predict student performance, with a stacking meta-model significantly enhancing predictive accuracy.
- The study employs SHAP to interpret model predictions, highlighting the importance of variables such as access to resources, critical thinking skills, and demographic factors.
- It demonstrates that factors like grade repetition, digital device ownership, and participation in additional Mathematics classes significantly impact performance.
- A state-of-the-art interactive dashboard was developed, integrating UMAP visualizations and SHAP analysis to facilitate decision-making for educators and policymakers. It provides actionable insights for addressing regional and demographic inequalities in student achievement, offering a framework for evidence-based policymaking.

The rest of this paper is organized as follows: Section II of the paper is a review of literature focusing on various determinants of students' achievements in Mathematics, such as teaching quality, gender, technology use, and availability of resources in homes. Section III depicts the approach taken, including the data processing, the adopted classification models, and the application of SHAP values for analysis. In Section IV, the results of the experiments are described, featuring the outcomes of the single classifiers and the meta-classifier in distinguishing student performance levels. Section V elucidates the results and their relevance to education. In contrast, Section VI re-evaluates the whole research and reiterates the power of integrating the ML models with the interactive dashboards for improving educational performance, and outlines future research opportunities.

II. RELATED WORK

This section analyses the different determinants of educational achievement with emphasis on Mathematics. It first considers the importance of teaching quality and what other strategies can bring about. Then, the section addresses the issue of gender in Mathematics throughout various settings. Technology integration as an influencing factor is discussed in the following subsection, where the drawbacks and aspects of technology engagement are also elaborated on and explained. Finally, it addresses the household educational resources, distinguishing parental engagement and the provision of capital as the most significant for children's achievements in Mathematics.

A. TEACHING QUALITY AND ITS IMPACT ON MATHEMATICS ACHIEVEMENT

An investigation in the Philippines scrutinized the interplay among socioeconomic status, a growth mindset, and proficiency in Mathematics and Science [21]. The outcomes indicated a positive association between a growth mindset and performance in these disciplines. Notably, the influence of a growth mindset on educational attainment was more marked in students hailing from economically disadvantaged backgrounds, underscoring the moderating role of socioeconomic status [22], [23]. The study also noted that students from lower socioeconomic strata are likelier to exhibit a diminished growth mindset than their more affluent counterparts, likely reflecting disparities in access to educational resources and opportunities.

B. TECHNOLOGY INTEGRATION AND ITS IMPACT ON MATHEMATICS ACHIEVEMENT

In examining the intersection of technology use and academic performance, particularly in Mathematics, recent studies offer nuanced insights into how digital tools influence learning outcomes. In [24], the authors, through their analysis in the PISA 2018 report, shed light on the complex relationship between technology usage and academic performance. Their research underscores the potential of technology to enhance learning when integrated thoughtfully into the educational process, suggesting that strategic application of digital resources can positively impact student achievement in Mathematics. Additionally, studies have unveiled a nuanced relationship between the employment of Information and Communication Technologies (ICT) and academic achievement in Mathematics [25], [26], [27]. Judicious use of ICT can bolster learning by facilitating practice, exploration, and access to online educational resources. Conversely, evidence suggests that excessive or misguided use of ICT might result in adverse academic outcomes [28], [29], [30]. The authors in [31] deepen into the effects of ICT use and related attitudes on students' Mathematics and Science performance. Their study, leveraging a decade of PISA surveys, highlights the importance of ICT access and quality use in educational settings. They argue that while ICT can be a powerful tool for educational engagement and improvement, its benefits are contingent upon students' attitudes towards technology and its application in learning Mathematics and Science [7], [32]. Furthermore, in [33], based on PISA 2018 data, employs multilevel structural equation analysis to understand the impact of educational ICT resources on student engagement and academic performance. Their findings point to a significant positive relationship between access to educational technology resources, student engagement, and academic performance in Mathematics. This suggests that the availability of digital tools, when coupled with effective engagement strategies, can substantially enhance students' learning experiences and outcomes in Mathematics.

C. GENDER GAP IN MATHEMATICS ACHIEVEMENT

There is a notable persistence of gender gaps in Mathematics achievement, with males typically surpassing females in various countries. An analysis of PISA data in [34] across 40 nations revealed substantial variation in this gap, with some countries reporting negligible gender differences and others noting females outperforming males. Further scrutiny of the 2018 PISA data suggested a narrowing gender gap in Mathematics performance, with females excelling in Mathematics in several regions. Despite this progress, disparities remain in selecting technical professions and confidence in Science, Technology, Engineering, and Mathematics (STEM) fields [35].

D. HOME EDUCATIONAL RESOURCES AND THEIR IMPACT ON MATHEMATICS ACHIEVEMENT

In the discourse on educational outcomes, particularly within the realm of Mathematics achievement, the availability of educational resources at home, such as books, stands out as a significant predictor of student performance. In [36], further into this aspect by examining the interplay between socioeconomic status (SES), migration background, non-cognitive abilities, and their collective impact on students' performance in PISA assessments for Reading and Mathematics [37], [38], [39]. The findings from this research underscore the importance of home educational resources, revealing that students with access to a broader range of books and educational materials at home tend to perform better in Mathematics. This correlation highlights the critical role of the home environment in shaping students' academic capabilities, suggesting that educational resources at home provide direct learning opportunities and reflect broader socioeconomic factors that contribute to academic success.

The review portrays the most valid factors affecting Mathematics success, such as teaching, gender, technology, and family support in education. Previous studies have investigated these areas, but it is clear that more work is needed to see how these aspects come together. In particular, the interaction of home resources with external ones, such as school quality, the role of digital resources, and the effect of attitudes towards STEM among different genders, have been sparse. In this regard, our methodology fills these gaps in the literature as it uses a multidimensional approach to unveil the multifactor nature of the determinants of achievement in Mathematics.

III. METHODOLOGY

In this section, we outline our methodological approach tailored to the analysis of the PISA 2022 dataset for Spain, detailing our rigorous data curation processes, the application of Uniform Manifold Approximation and Projection (UMAP) for thoughtful dimensionality reduction [40], and the deployment of various binary classification models aimed at decoding the diverse academic performance levels of students. Central to our strategy is the delicate balance between computational precision and the interpretability of our findings, ensuring that the insights garnered are statistically robust and pedagogically meaningful. The overarching goal of our methodical workflow is to convert a comprehensive set of educational data into actionable insights that can inform policy and pedagogy, thereby catalyzing the improvement of student outcomes within the academic landscape.

TABLE 1. The level of proficiency in Mathematics skills of the cohort surveyed, breaking it down by score range, descriptive level, how many individuals achieved that level, and a description of such levels. This breakdown level is significant because it gives the overall picture of the mathematical skills presented in the sample and the distribution of the different proficiency levels, thus providing a deep detail regarding mathematical skills within the sampled population.

Score Ranges	Levels	Number of Samples	Categories
score ≤ 295	Level 1c	4396	Low
$295 <$ score ≤ 358	Level 1b	1709	Low
$358 <$ score ≤ 420	Level 1a	168	Low
$420 <$ score ≤ 482	Level 2	7681	Low
$482 <$ score ≤ 545	Level 3	8442	Medium
$545 <$ score ≤ 607	Level 4	5201	Medium
$607 <$ score ≤ 669	Level 5	1490	High
score > 669	Level 6	195	High

A. DATA CURATION FROM PISA 2022 FOR SPAIN

The dataset employed in this investigation originates from the latest 2022 PISA assessment, focusing on student performance within Spain. Provided by the Organisation for Economic Co-operation and Development (OECD), the raw dataset comprised data from 30,800 students. A foundational aspect of our data curation involved the removal of entries with excessive missing values, a decision guided by the principle of maintaining rigorous data quality for analysis.

Out of the available rows with all necessary data, variables were chosen that served the purpose of the study and covered fundamental educational, demographic, and resource-related aspects. Any variables that were substantially blank or not much linked to the study objectives were removed. This approach resulted in a dataset that was both wide-ranging and suitable for producing trustworthy research outcomes.

As Figure 1 illustrates, the dataset exhibited many rows with many missing values. The red line in the histogram, representing a 95% confidence threshold, delineates the cutoff for rows with nine or more missing values (columns with no values). Rows exceeding this threshold were excluded from further analysis. This preemptive data filtering resulted in a condensed dataset of 29,282 Spanish students, reducing 1,518 entries from the original sample. Such a stringent criterion for data inclusion ensures the analytical robustness of our study by mitigating the potential bias that could arise from a high incidence of incomplete data. The remaining missing values, whose rows can be a maximum of nine missing values, have been replaced by the mean value of the entire dataset.

The variables maintained for the study, as enumerated in Tables 5 and 6, were selected with an emphasis on constructing a dataset that is both comprehensive and conducive to dependable research findings. This histogram in Figure 1 and the data cleaning threshold it illustrates testify to the meticulous approach adopted in preparing the data for this study.

Justification for the Three-Category Classification System: It is necessary to explain why we moved from a seven-tiered structure to a simplified three-category classification system as we go deeper and deeper into our methodological framework. This consideration was primarily because it was deemed necessary to simplify aspects of the analysis as we contend with the complexities of large educational datasets. The logical regrouping of the original seven levels into three broader categories aligns with some research traditions in educational studies. Works by the authors in [16] and [17] support this, which highlight the benefit of such simplification as directing the focus on essential patterns and conclusions while maintaining the analytic quality of the study. So, in our study, this strategy allows us to address significant aspects and provides well-supported and comprehensible results within the educational context.

This type of systematic reduction reveals simple regularities and factors concerning academic attainments and progressions. In the preprocessing step, the dummy binary variables were converted into one-hot encoding, one method used to transform categorical variables into numerical values. Thus, our data could be processed more effectively with ML methods.

One-hot encoding does not permit the inadvertent ordering of more than two removal levels within the utopian concept matrix categories to guarantee accurate and detailed data representation (Table 1 shows the distribution of categorization levels of mathematical skill after obtaining them). Note that the labels determining the level of proficiency in Mathematics were left in the original form, as this was the case with the most categorical variables subjected to one-hot encoding. Because of this, it was indicated how the integrity of the analytical results should be preserved and, therefore, the disorder included. Because such inscriptions label distinct and discrete classes, it was unnecessary and, thus, in the light of the purpose of this study, inappropriate to use one-hot encoding on these particular variables.

B. OVERVIEW OF BINARY CLASSIFICATION MODELS

In pursuit of this aim, we considered eight different binary classification models and used them on the data containing the three levels of Mathematics scores: Low, Medium, and High. The models were trained using the stratified K-fold approach, which was enhanced by an undersampling technique to eliminate the imbalance in data allocation within each fold and ensure that each model got adequate samples. The next rounds of optimization included various hyperparameter tuning and employing grid search to find the best model among the pool of trained candidates. Here, we summarize the models we investigated.

Logistic Regression (LR) is a fundamental model for binary classification, predicting the probability that a given input belongs to a particular category. Consider a collection of p independent variables denoted by the vector $\boldsymbol{x} = (x_1, x_2, \ldots, x_p)$. The model estimates probabilities using a logistic function, which is expressed as:

$$p(y = 1|\boldsymbol{x}) = \frac{1}{1 + e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2+\cdots+\beta_p x_p)}}, \tag{1}$$

where $p(y = 1|\boldsymbol{x})$ is the probability of the input $\boldsymbol{x}$ being classified as class 1, β_0 is the intercept, and β_1 is the

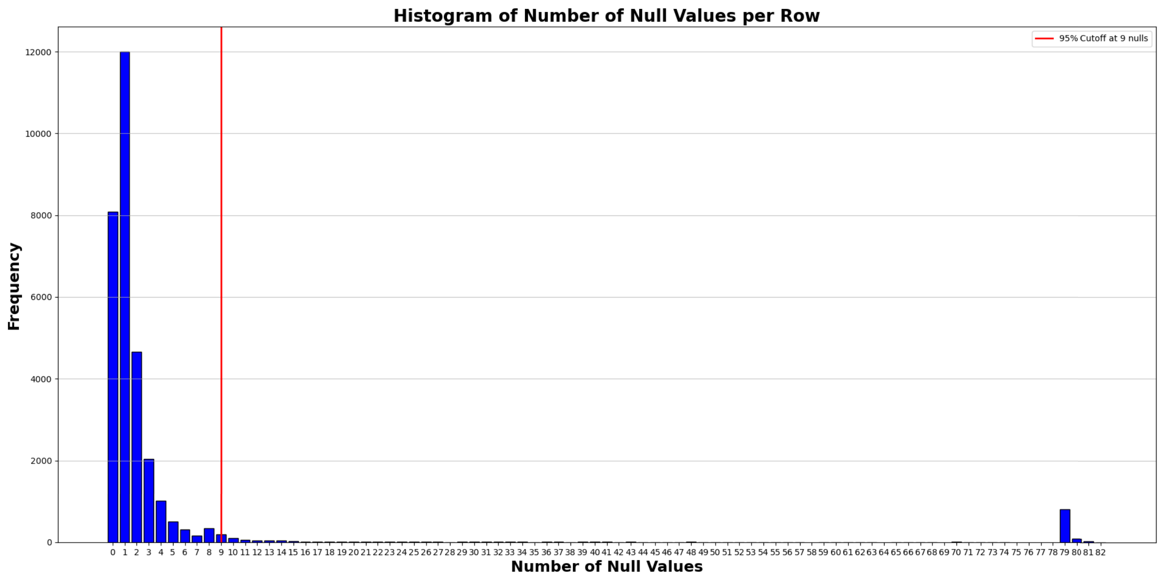


FIGURE 1. The number of missing values in the data set provided by a histogram is likely to be associated with students. The case is in red and represents the cut value. If the number of empty columns exceeds this number or 'cut', all columns in this range are considered almost empty. Each bar denotes the number of students with the corresponding number of empty columns. The red line assists in determining the number of unoccupied columns, which is intentional and has a more sustained use than the employment in data cleaning.

coefficient for the input feature x. This model is favored for its simplicity and interpretability [41], [42].

Support Vector Machines (SVM) separate various classes by generating a hyperplane in several dimensions. The optimal hyperplane is the one furthest away from the closest points of two classes, referred to as the support vector. To separate such data, non-linear boundaries SVM relies on kernel functions, which project the input data into higher dimensions. Therefore, the decision function is mathematically expressed as follows:

$$f(\boldsymbol{x}) = \text{sign}\left(\sum_{i=1}^{n} \alpha_i y_i K(\boldsymbol{x}_i, \boldsymbol{x}) + b\right), \tag{2}$$

where $K(\boldsymbol{x}_i, \boldsymbol{x})$ is the kernel function, y_i are the labels, α_i are the Lagrange multipliers, and b is the bias [41], [43], [44].

Decision Trees, as the name suggests, cause the regression or the classification to take the form of a tree by dividing the datasets into smaller portions according to the input feature values. In this case, the aim is to ensure that the leaves are monolithic regarding their class. All decisions in the tree that is built at each stage are based on a feature that optimally separates the current class and its siblings according to some beliefs of the classifier, such as information gain:

$$IG(\boldsymbol{D}_p, \boldsymbol{f}) = I(\boldsymbol{D}_p) - \sum_{j=1}^{m} \frac{N_j}{N_p} I(\boldsymbol{D}_j), \tag{3}$$

where $IG(\boldsymbol{D}_p, \boldsymbol{f})$ is the information gain of parent dataset $\boldsymbol{D}_p$ by feature $\boldsymbol{f}$, I is the impurity measure, N_p is the number of samples in the parent set, and N_j is the number of samples in the j-th subset [41], [45].

Random Forest upon Decision Trees by increasing the power of the predictive algorithm to bring an ensemble of trees to bear the predictive task with their aggregates used. Each tree is trained on a random subset of the data, and the final classification is typically determined by majority vote: $\hat{y} = \text{mode}\{y_1, y_2, \ldots, y_n\}$, where $\hat{\boldsymbol{y}}$ is the predicted class and y_i are the predictions of the ith tree in the forest [46].

Gradient Boosting (GB) constructs a weak model such as a decision tree and continues adding them so that the later ones address the mistake of the earlier trees. Each model is built using the commissioned framework in favor of each particular model, as given below:

$$F_m(\boldsymbol{x}) = F_{m-1}(\boldsymbol{x}) + \rho_m h_m(\boldsymbol{x}), \tag{4}$$

where $F_m(\cdot)$ is the model at iteration m, $h_m(\cdot)$ is the weak learner, and ρ_m is the weight of $h_m(x)$ [47].

XGBoost is an optimized distributed GB library designed to be highly efficient, flexible, and portable. It performs the functions of many ML algorithms within the parameters of GB with improvements for speed and performance [48].

A *Multilayer Perceptron (MLP)* consists of any feedforward architecture with at least three layers: the input layer, the hidden layer, and the output layer. Each neuron output is a nonlinear function of the weighted summation of its inputs:

$$y = \sigma\left(\sum_{i=1}^{n} w_i x_i + b\right), \tag{5}$$

where σ is the activation function, w_i are the weights, x_i are the inputs, and b is the bias. MLPs can model complex relationships between inputs and outputs, making them suitable for various applications, including classification and regression tasks [41], [49].

LightGBM is a GB framework that uses tree learning algorithms. The system is optimized for speed and efficiency, emphasizing big data. LightGBM advances the performance of the standard techniques of GB by implementing a new form of tree-building strategy called the Gradient One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB), which significantly helps in reducing the cost of computations when working with significant datasets where GOSS retains cases with high gradient values and randomly takes cases with less gradient value while EFB groups no overlapping features to cut down the number of features [50].

These models were rigorously tested to ascertain their effectiveness in binary classification tasks across different mathematical proficiency levels, employing a structured approach to ensure fairness and reliability in performance evaluation.

C. STACKING META-MODEL APPROACH

Stacking, introduced by Wolpert in [18], is a significant contribution towards advancing the field of ML as it presents a new way of combining model outputs based on a meta-model whose parameters are trained using the outputs from several base models. Such an approach, in essence, reduces the generalization error, greatly facilitating the emergence of other forms of stacking known as stacking meta-models or simply meta-stacking. The term meta-stacking is an enhancement of the initial term of stacking since this involves combining one or more base learners along with one or more meta-models, thus demonstrating the relevance of Wolpert's work in the recent developments in ensemble methods.

Upon assessing eight binary classification models, the next step is introducing a stacking meta-model strategy. This advanced ensemble technique maximizes the output of the team of models as individuals deriving their own merits from various models. The main point is that a particular model, called meta-modeling, is built to consolidate the multiple model predictions and consolidate them into one prediction [18], [19], [51].

Recent studies in the field of educational data mining have started to employ stacking methods in solving more complicated predictive issues involving datasets from PISA. For example, in [51], the authors showed that when student performance is predicted, stacking improves the accuracy of the model by making use of several base models simultaneously [51]. Such developments indicate the increasing importance of stacking in education concerning data sets like the PISA, for which complex interactions between the variables of interest are present that would need elaborate modeling techniques to capture.

In the conducted stage of our methodology, we apply the advanced meta-model construction known as stacking, in which base learners of varying structures are combined to obtain more accuracy than any single base model can provide. This has mainly been achieved through a meta-model whereby instead of one model making a prediction, many models are used to make many predictions. Then, all these predictions are averaged to create the final prediction.

Exhaustive Hyperparameter Search for Base Models. As a first step, individual exhaustive hyperparameter searches are performed on all eight binary classifications. This systematic hyperparameter search encompasses a search over the hyperparameters of each model and helps to identify the best combination for its construction. This wide-scope search aims to ensure that the performance of each model is improved one at a time. In the end, the best-performing model of each category is taken forward for incorporation in the stacking architecture.

Selection and Training of the Meta-Model. When the base models have been individually optimized, a stacking strategy is then used to prepare the base model predictions for the input features of the meta-model. In this case, since we have two level architectures, we have settled on LR as the meta-model for our study due to its effectiveness in binary classification tasks, ease of use and understanding, and also the flat nature of the decision boundary, which is optimal for combining the outputs from several different models.

This is to say that there are also other steps for hyperparameter optimization in the meta-model training process. Still, they are pretty narrowed down since there is an explicit LR meta-model. This step ensures that the best-generating features from the base models are integrated into the LR model to ensure maximum efficiency of the stacking meta-model.

Implementation Details. For all components of the stacking framework, that is, the base models and the meta-model, the cross-validation approach is stratified K-fold. It also helps care for the baseline models and the meta-model learning on the unseen data and the proficiency distribution over folds. The exhaustive hyperparameter search and training stage for the LR meta-model is done by understanding the overfitting factors and the strain that may consume too much time in computation.

Evaluation and Optimization. An array of evaluation metrics was employed on the stacking meta-model evaluation to assess the stacking performance. Optimization activities would then be directed to LR meta-model hyperparameters after pursuing all base model optimization and search forms. This strategy guarantees that the stacking technique uses the separate components' predictive capabilities and corrects all their predictions to achieve maximal prediction accuracy.

D. STRATIFIED K-FOLD CROSS-VALIDATION FOR BINARY MODEL TRAINING

Stratified K-Fold cross-validation is another necessary and significant method, especially for classifying models where datasets have class imbalances. In this method, there is a lot of assurance that classes will be evenly represented in all the separate buckets, which is optimal for evaluating the performance of the models [52], [53].

In this work, the dataset was divided into 80 % train and 20 % test (considering the Pareto rule), where this train set was trained using the k-fold stratified method, with a $k = 5$. With the model already trained, it was validated using the evaluation metrics (discussed in the following subsection) with the remaining 20 % for the test [54].

As mentioned, this process is carried out five times, whereby in each fold, only one fold is used as the validation fold in one iteration, so all the folds are used precisely once. One of the essential characteristics of the optimization approach is that all the models are trained more or less from the beginning in each iteration to eliminate the effect of already trained models. The resulting metrics obtained based on each of the folds are then pooled together, which brings about the most thorough evaluation of a particular model performance [55].

E. EVALUATION METRICS

This study employs a suite of six essential metrics for assessing the performance across all models, which are the following ones: Accuracy (ACC), Recall (RC), Precision (PR), Specificity (SP), F1 Score (F1S), and Area Under Curve (AUC) [56]. ACC offers a performance evaluation for the initial metric by denoting the ratio of accurately predicted instances to the overall number of predictions. The ACC is determined by the formula:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}, \tag{6}$$

where TP denotes the true positive rate, TN is the true negative rate, FP is the false positive rate, and FN is the false negative rate. RC is indicative of the model efficacy in accurately identifying all positive instances, proving crucial in scenarios where FNs are deemed more detrimental than FPs. It is computed as:

$$RC = \frac{TP}{TP + FN}. \tag{7}$$

PR quantifies the ratio of accurately identified positive instances (TP) to all instances classified as positive ($TP+FP$), expressed as:

$$PR = \frac{TP}{TP + FP}. \tag{8}$$

SP, conversely, assesses the ratio of accurately identified negative instances (TN) out of all instances classified as negative ($TN + FP$), as delineated by:

$$SP = \frac{TN}{TN + FP}. \tag{9}$$

The F1S is a harmonic mean of PR and RC, balancing the model accuracy and completeness. It is calculated via:

$$F1S = \frac{2 \cdot (PR \cdot RC)}{PR + RC}. \tag{10}$$

Lastly, AUC reflects the integral of the Receiver Operating Characteristic (ROC) curve, encompassing both the TP and FP rates. The AUC calculation is as follows:

$$AUC = \int_0^1 ROC(t)dt, \tag{11}$$

where an AUC of 1 signifies an exemplary classification model and t represents the threshold used to classify the predicted probabilities into positive and negative classes. In contrast, a score of 0.5 indicates a performance equivalent to random guessing [57].

F. INTERPRETABILITY IN MACHINE LEARNING THROUGH SHAPLEY VALUES

There are a lot of subfields of cooperative game theory that resort to the so-called Shapley values, such as participation in a cooperative game will return average proportional rewards [58]. Translated into ML, these participants are treated as features or attributes of the model. Transforming the coalition into the predictive model implies a task to be carried out. Hence, the Shapley vector interprets the importance of all the attributes in predicting the outcome of the ML model [59]. In particular, given a query point, the Shapley value of a feature describes how the predicted value (regression) or class scores (classification) will change based on that feature inclusion. More particularly, the contribution of a feature is determined by the amount of variation in prediction from the average prediction of the model at a specific query point due to that feature. This view, one can suppose, assigns the Shapley value regarding the i feature at x query to the value function v_x as

$$\varphi_i(v_x) = \frac{1}{M} \sum_{S \subseteq M_s \setminus \{i\}} \frac{v_x(S \cup \{i\}) - v_x(S)}{\binom{M-1}{|S|}}, \tag{12}$$

where M is the total number of the features, M_s is the family set of all the features, $|S|$ refers to cardinality of the set S, and $v_x(S)$ is the mean profit variation at point x due to S features.

The SHAP Python package offers a set of interventional algorithms to compute these Shapley values as well [20]. The algorithms work by determining the value function of some attributes defined at a point of interest within a query as the expected prediction given an intervention distribution which exposes all other attributes not within the set of attributes being referenced:

$$v_x(S) = E_D[f(x_S, X_{S^c})], \tag{13}$$

where x_S represents the values of the features in set S at the query point, and X_{S^c} those not in S. Assuming features are independent, the value function $v_x(S)$ is approximated using data X as samples from the intervention distribution D for S^c:

$$v_x(S) = E_D[f(x_S, X_{S^c})] \approx \frac{1}{N} \sum_{j=1}^{N} f(x_S, (X_{S^c})_j), \tag{14}$$

with N representing the sample size, and $(X_{S^c})_j$ the values for S^c in the j-th observation.

Interventional algorithms, while computationally more efficient, presume feature independence and rely on samples from outside the distribution, potentially leading to unrealistic observations [60]. Key SHAP interventional algorithms include: (1) *Kernel SHAP*, employing a kernel approximation for Shapley value estimation, suits high-dimensional spaces albeit with higher computational demands [20]; (2) *Linear SHAP* offering an analytical computation of Shapley values for linear models, enhancing computational efficiency compared to Kernel SHAP [20]; and (3) *Tree SHAP*, tailored for tree-based models like Decision Trees, Random Forests, and GB algorithms, excelling in computational efficiency and explicitly accommodates feature interactions [61].

G. DIMENSIONALITY REDUCTION FOR PISA DATA ANALYSIS

In this research section, we describe a UMAP approach concerned with efficiently reducing the dimensionality of PISA data. Unlike existing techniques, which do not incorporate the class labels while performing dimensionality reduction, UMAP facilitates the preservation of relevant geometrical features in a lower dimensional space, which concerns the class separability [40].

1) DATA PREPROCESSING AND PARTITIONING

As the first step in our processes, the preprocessing stage involves eliminating attributes that do not contribute significantly to analyzing the PISA students' performances. This process sharpens the model focus on significant predictors. A split of 60/40 of the data serves as sufficient training data for the model and adequate testing data for the validation of the model. To change such characteristic features values to a common scale *StandardScaler* was employed and after this, better results were expected in subsequent training phases [62].

2) UMAP CONFIGURATION AND TRAINING

The UMAP model, configured to reduce the degree of latitude of data into a lower dimensional three-dimensional latent space, has its parameters set on how to depict local and overall data structures depending on the set hyperparameters, *n_neighbors=50* for the local neighborhood size of the data, *n_components=3* to visualize the data in a three-dimensional space, and *min_dist=0.1* to enable the low level embedding to spread out as much as possible. This transformation of the PISA data aims to pack the different indicators behind the essay into the low dimension while ensuring that the community-based segregation of data as per the labels is retained.

3) INTERACTIVE DASHBOARD FOR PISA DATA ANALYSIS

A solid interactive dashboard has been constructed using the latent dimensions of the PISA students' data supplied by the UMAP. This solution is built using *Dash* framework [63], enabling a fast and flexible web interface for conceiving Python applications. To deliver the interactive dashboard and insightful visual explorations of the multi-dimensional PISA data, the dashboard is enhanced with *Plotly* graphing library [64].

4) TECHNOLOGIES AND FRAMEWORKS

The *Dash* framework is employed due to its ability to integrate the Python programming language in converting, for instance, sophisticated data analysis into effective web-based interactive designs. Enriching this experience is *Plotly* that provides various interactive graphics [63], [65]. All these are the central part of a dashboard that encourages a detailed investigation of the complex PISA database.

5) DASHBOARD FEATURES

The dashboard possesses the following important aspects: (1) *3D UMAP Visualization*, there is an interactive *scatterplot* in 3d where many students with many attributes are presented in an interactive three-dimensional scatter plot; (2) *SHAP Value Visualization*, which attempts to answer the question on what impacted which particular student's performance predictive features; and, (3) *Responsive Design*, to retain the users across the various devices that may be used to access the dashboard. For detailed UMAP-derived insights from the community, the user is provided with interactivity features like drop-down selectors and clickable plot points, enabling the female guinea pig to be less specific.

H. RESEARCH WORKFLOW STRATEGY

Effectively analyzing the PISA 2022 data starts with selective data cleaning processes, as evidenced by Figure 2. Increasing biases creates a series of challenges because it sets several limitations attached to how datasets should be pre-processed and organized. After preprocessing, the data is split into training and test sets, a critical process in any model-building exercise and performance assessment to avoid overfitting.

To address the class imbalance issue in the PISA dataset, the present study employs a stratified 5-fold cross-validation method in conjunction with random undersampling techniques during the model training. This method ensures that each fold has an equal number of instances belonging to each class, thereby promoting fairness and ACC of the developed model. The analysis continues rather rapidly with all sorts of fancy workable ML models, LightGBM, MLP, LR, Decision Trees, Random Forests, SVM, XGBoost, and many other hard-core ML modeling, with grid searches for parameters optimization that come along in hand. Such modeling efforts are comprehensive to enhance the capability of the model being selected for binary classification tasks, focusing on increasing the correct classifications.

To provide visualization and clustering of high-dimensional data, UMAP, a popular dimensionality reduction technique, is applied to the data. Regarding dimensionality reduction, SHAP values are computed on the original variables in order to give a local and global view of model predictions. This kind of workflow guarantees that while the latent space offers a broad view and understanding of the data, SHAP values unveil the specificity of each feature. The degree of correlation between these two aspects of the framework is well illustrated in the interactive dashboard (Figure 7). When a specific spot is clicked on the UMAP projection, SHAP values corresponding to that student are revealed to the user, directly linking visualization and explanation.

Model effectiveness is gauged in numerous metrics, such as ACC, RC, PR, SP, F1S, and AUC of the ROC curve. These arguments support a well-rounded appraisal of model performance in terms of the errors that are quantitatively measured in predicting some parameters.

After such benches for each model, we propose a new meta-model strategy to align all the former models optimally grouped around their parameters. Thus, this meta-model treats the parameters of the regression models as a structure upon which an ensemble of diverse algorithms that has been defined before is built. It is this integrated approach where the study seeks to exploit the predictive power of every individual model in a case that interacts with a few key parameters as hyper-parameter tuning has been previously optimized in grid search attempts. This meta-analysis phase consists of the sequential combination and evaluation of the hyperparameters used in the ensemble to yield one optimal model configuration. Last, in this process, the most efficient meta-model emerges, which marks one of the progress of the analysis in the study.

The model evaluation stage ends with a deep understanding of the last section, which deals with SHAP-based interpretability and the meta-model selected as the best-performing and the least variable classifier. This aspect of SHAP analysis refers to the extent to which every single detail in the judgment of the model contributes and, therefore, seeks to explain how the meta-model is built and its decisions. This SHAP analysis is focused on the contribution of the PISA 2022 dataset within the multifaceted and explanatory models.

Regarding variables, the feature selection strategy highlights factors that positively impact education. This is done through SHAP interpretability, which helps locate these characteristics and rank them not only by their statistical importance but their instructional value as well. The aim, therefore, is to identify patterns and variables that significantly impact student achievement in the PISA assessment so that beneficial actions may be taken by those responsible for education. 'Analysis Application' is another phase of the analytical approach, which considers the practical implication of the results. This consumes the educational data, so only the most informative features remain. In addition, since the purpose of local diagnostics is to explain predictions for each observation, this risk involves more focused educational objectives.

IV. EXPERIMENTS AND RESULTS

This section presents the findings from evaluating eight binary classification models across three distinct scenarios of student performance in Mathematics: Low vs. High, Medium vs. High, and Low vs. Medium. Furthermore, it discusses the results obtained from the meta-model and delves into the variables deemed most significant by the SHAP analysis, highlighting their impact on students' mathematical performance.

A. ANALYSIS OF BINARY CLASSIFICATION MODELS AND META-MODEL FRAMEWORKS

In the present analysis, we examine and assess the operating outcomes of numerous ML models used to classify different categories of student performance across Low-Medium, High-Medium, and Low-High divisions. Table 3 contains summary performance results for each model concerning different cases in the investigation.

Regarding the Low-High categorization, the models under evaluation were LR, SVM, Decision Tree, Random Forest, GB, XGBoost, MLP, and LightGBM. The one exception in performance is the meta-model, which is an ensemble model with LR as a base and performed better than other models on all metrics of interest, especially its F1S, PR, SP, and AUC. These results prove that the meta-model can distinguish the Low from the High-performance segment. The performance is due to the combination of various models that work in cooperation with each other rather than individual approaches.

The trends in model performance are similar for the High and Medium-Low categories. The difference is that in the Medium-High category, once again, the meta-model beats the individual models, reiterating the success of ensemble approaches to complex classification challenges. On the other hand, the particular model performances in the Low-Medium categorization bring out the unique abilities of the Random Forest and the meta-model, which once again summarily perform best in most metrics.

Hyperparameter tuning proves significant during model optimization and warrants mention. Table 2 shows the predetermined hyperparameter configurations, which proved reasonably optimal in three scenarios. For instance, hyperparameters like *learning_rate*, *n_estimators*, or *max_depth* are adjusted for each model by the specificities of the particular experimental conditions of that specific model.

Low and High-Performance in Mathematics: Factors that make it possible to forecast success in the working of students within Mathematics are still being sought after, and thus, synthesizing these along with the interpretability of the model remains critical. The SHAP approach guides the evaluation of the role of each feature within model output, therefore improving the understanding of how models make decisions. Looking at Figure 3, we can see a significant difference between the SHAP values assigned to the student's

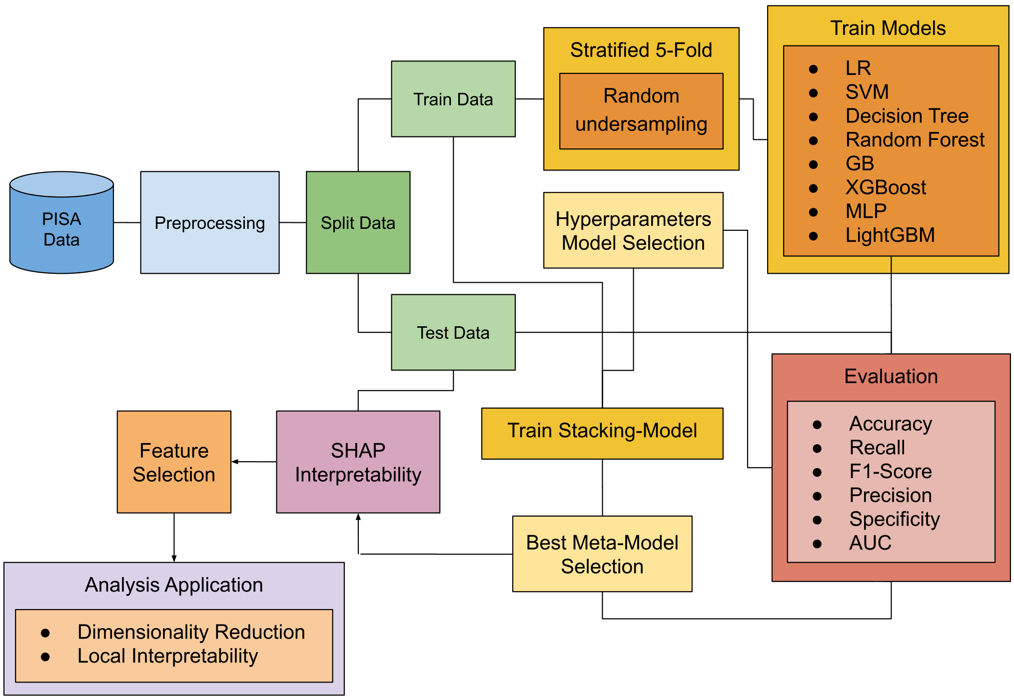


FIGURE 2. PISA 2022 data analysis starting with: (1) the preprocessing of data and splitting it into training and test datasets; (2) resolving problems resulting from class imbalance by conducting stratified 5-fold cross-validation with random undersampling; (3) training of models consisting of different algorithms such as LR, SVM, Decision tree, Random Forest, GB, XGBoost, MLP and Light GBM with hyperparameter optimization; (4) evaluation of the model performance on the ACC, RC, F1S, PR, SP, and AUC metrics; (5) performing a selection of the most applicable meta-models for analysis; (6) increasing explanatory power using SHAP; (7) making feature selection; and (8) performing the analysis using these features.

demographic features compared to high-resource students. More interesting is the variable 'REPEAT' (which explains whether a student is a repeater), which is one of the strong predictors of low performance in Mathematics. This is also shown in the swarm plot in Figure 3 (a), which shows that the loading of 'REPEAT_YES' on the negative values of SHAP indicates that it is more associated with students with a Low level in Mathematics. Also, the bar chart presented in Figure 3 (b) describes the average effect of most active features on students with the highest performance level. As in Figure 3 (c), the heat map management of SHAP values is shown supported by the position of the values across the dataset.

1) REPEATING STUDENT (REPEAT)

The impact of the 'REPEAT_YES' variable is quite pronounced for the Low-performance levels in Mathematics, as can be seen by the deep red colors in the swarm plot (Figure 3 (a)). This means that students repeating a particular grade tend to do very inadequately in Mathematics, which further emphasizes the importance of this variable.

2) DIGITAL DEVICES WITH SCREENS AT HOME (ST253Q01JA0)

The variable quantifies how many digital devices with screens (television, computer, among others.) are held by a student in their home environment, with a value of 8 representing that more than ten devices are available. Looking at the color gradients in Figure 3, especially in segment 3 (c), it can be concluded that it is prevalent or positively correlated with performance (more than ten devices). However, these cases are not displayed as red samples. On the contrary, this could mean that having an extremely high number of devices is not necessarily an advantageous element for a child's performance in Mathematics, as they could be a source of distraction. Thus, too much freedom could be harmful to education. It is essential to find the optimal ratio of the presence of technological resources and their directed use for educational needs.

3) NUMBER OF BOOKS AT HOME (ST255Q01JA)

This variable interpretation measures the number of books that are likely present in the student's home environment, and two (2) books mean the number of books from 11 to 25. Purple tones surging at the lower ends of the SHAP values suggest that those moderate numbers of books have less effect on predicting very high performance in Mathematics. This bookless scenario, while plausible, does not definitively erase the role of books at home; rather, it probably indicates that, beyond a certain point, the number of books is associated more with better academic performance than with a lack of books. This observation could imply that a lower limit of books may be useless and that a sufficient number of books (say 20 or more) will promote better learning.

TABLE 2. Best hyperparameter values were carefully chosen based on expected performance from the different experiment scenarios. This table classifies the parameters that were most optimal for the various models across three different experimental scenarios: Low-High, Medium-High, and Low-Medium.

Model	Hyperparameter	Low-High	Medium-High	Low-Medium
LR	'penalty': l1, l2 'C': 1, 10, 100	l2 10	l1 1	l2 10
SVM	'kernel': linear, rbf 'C': 1, 10, 100	rbf 10	rbf 10	rbf 100
Decision Tree	'criterion': gini, entropy 'max_depth': None, 5, 10	gini None	entropy None	entropy None
Random Forest	'n_estimators': 50, 100, 200 'max_depth': None, 5	100 None	200 None	200 None
GB	'n_estimators': 100, 200, 300 'learning_rate': 0.1, 0.01	300 0.1	300 0.1	300 0.1
XGBoost	'n_estimators': 100, 200, 300 'learning_rate': 0.1, 0.01	300 0.1	300 0.1	300 0.1
MLP	'hidden_layer_sizes': (100,), (100, 50, 25) 'activation': relu, tanh, logistic	(100,) tanh	(100,) tanh	(100,) tanh
LightGBM	'n_estimators': 50, 100, 200 'learning_rate': 0.1, 0.01	200 0.1	200 0.1	200 0.1
Meta-Model: LR	'C': 0.1, 1, 10, 100 'solver': liblinear, saga 'penalty': l1, l2	100 saga l1	100 liblinear l1	0.1 saga l2

4) PARTICIPATION IN EXTRA MATHEMATICS CLASSES (ST297Q09TA)

Considering the distribution of SHAP values, where participation does not seek out additional Mathematics classes (value 0) results in red hues in the Low-performance category, which implies that participation in extra Mathematics classes is an effective predictor of performing inadequately. On the other hand, blue hues may suggest that letting go of additional Mathematics classes might not negatively influence high-performing students who may find themselves in high school already without the need for extra classes. Thus, SHAP analysis is significant in contextualizing students' performance relative to their engagement with such factors and the attainment of Mathematics. It shows that repeating grades, resources available to children at home such as books and digital devices, and the variable engaging in voluntary Mathematics classes work interactively together with how students perform. Particularly, the variable 'REPEAT_YES' stands out as one the most important predictors of the variables discussed above, particularly for students with difficulty in Mathematics necessitating this group's targeted educational strategies.

Medium and High-Performance Study in Mathematics: Looking at the SHAP values of students with High and Medium ranking in Mathematics, we can briskly analyze the importance of each feature depending upon the opacity as presented in Figure 4. The depth of the color demonstrates the strength of each variable regarding the model expectation, with red being the most positive influence and blue being either mild or negative.

5) DIGITAL DEVICES WITH SCREENS AT HOME (ST253Q01JA0)

The SHAP values respond to this variable by adding a greater number of digital devices (8 for more than ten devices) positively correlates to better levels of performance in Mathematics seen in the red shades of the High-performance category of Figure 4 a. This implies that in the case of High and Medium achievers, better range of digital resources may be utilized in an effective cultural context for their educational goals of acquisition constructing achievement.

6) GENDER OF THE STUDENT (ST004D01T)

A variable intended for the measurement of the gender of a student and emphasizing 'Boy' displays a positive effect on the upper Mathematics performance than the Medium level tensions above red in the High-performance category and positioning of the variable in Figure 4. The disparity could perhaps be attributed to gender-based learning preferences.

7) STUDY OR DO HOMEWORK BEFORE GOING TO CLASS (ST294Q02JA0)

The SHAP module strongly confirmed the main conclusion. It showed that the probability of regularly studying or doing homework before class increases mathematical performance. Here, suppose the red shades appear prominently for the high values of this variable. In that case, it implies that regular practice of such activities before class contributes positively towards the students' performance.

TABLE 3. A comparative summary of performance measures of different ML models within three study cases: Low-Medium, High-Medium, and Low-High. These measures provide information about the classification capabilities of the models about the different levels of student performances attained.

	Low-High					
Model	**ACC**	**RC**	**F1S**	**PR**	**SP**	**AUC**
LR	0.8802 ± 0.0071	0.9036 ± 0.0175	0.6187 ± 0.0156	0.4706 ± 0.0175	0.8774 ± 0.0082	0.9557 ± 0.0042
SVM	0.9005 ± 0.0062	0.9334 ± 0.0145	0.6686 ± 0.0166	0.5212 ± 0.0194	0.8966 ± 0.0069	0.9717 ± 0.0038
Decision Tree	0.8349 ± 0.0097	0.8902 ± 0.0199	0.5369 ± 0.0180	0.3846 ± 0.0172	0.8282 ± 0.0108	0.8592 ± 0.0109
Random Forest	0.8965 ± 0.0064	0.9315 ± 0.0127	0.6594 ± 0.0158	0.5106 ± 0.0180	0.8923 ± 0.0072	0.9737 ± 0.0036
GB	0.8912 ± 0.0074	$\mathbf{0.9370 \pm 0.0135}$	0.6494 ± 0.0178	0.4972 ± 0.0197	0.8857 ± 0.0081	0.9670 ± 0.0039
XGBoost	0.8984 ± 0.0071	0.9366 ± 0.0148	0.6647 ± 0.0184	0.5156 ± 0.0217	0.8938 ± 0.0081	0.9709 ± 0.0033
MLP	0.9019 ± 0.0077	0.9332 ± 0.0150	0.6719 ± 0.0193	0.5253 ± 0.0231	0.8982 ± 0.0088	0.9680 ± 0.0042
LightGBM	0.8986 ± 0.0072	0.9361 ± 0.0144	0.6651 ± 0.0177	0.5162 ± 0.0211	0.8941 ± 0.0083	0.9725 ± 0.0041
Meta-Model: LR	$\mathbf{0.9157 \pm 0.0066}$	0.9330 ± 0.0158	$\mathbf{0.7042 \pm 0.0184}$	$\mathbf{0.5659 \pm 0.0229}$	$\mathbf{0.9136 \pm 0.0076}$	$\mathbf{0.9766 \pm 0.0040}$

	Medium-High					
Model	**ACC**	**RC**	**F1S**	**PR**	**SP**	**AUC**
LR	0.7057 ± 0.0107	0.7489 ± 0.0263	0.3586 ± 0.0143	0.2358 ± 0.0106	0.7003 ± 0.0114	0.7956 ± 0.0129
SVM	0.7554 ± 0.0100	0.8215 ± 0.0208	0.4247 ± 0.0156	0.2865 ± 0.0128	0.7473 ± 0.0109	0.8688 ± 0.0109
Decision Tree	0.6990 ± 0.0126	0.7899 ± 0.0243	0.3658 ± 0.0152	0.2381 ± 0.0115	0.6878 ± 0.0141	0.7388 ± 0.0135
Random Forest	0.8000 ± 0.0089	0.8214 ± 0.0245	0.4743 ± 0.0165	0.3336 ± 0.0145	0.7974 ± 0.0102	0.9013 ± 0.0110
GB	0.7361 ± 0.0101	0.8022 ± 0.0211	0.4004 ± 0.0133	0.2669 ± 0.0106	0.7279 ± 0.0113	0.8369 ± 0.0115
XGBoost	0.7710 ± 0.0103	0.8297 ± 0.0213	0.4432 ± 0.0150	0.3025 ± 0.0130	0.7637 ± 0.0119	0.8749 ± 0.0108
MLP	0.7603 ± 0.0148	0.8303 ± 0.0214	0.4325 ± 0.0157	0.2926 ± 0.0141	0.7517 ± 0.0177	0.8612 ± 0.0103
LightGBM	0.7731 ± 0.0092	$\mathbf{0.8326 \pm 0.0212}$	0.4463 ± 0.0147	0.3050 ± 0.0125	0.7657 ± 0.0104	0.8782 ± 0.0101
Meta-Model: LR	$\mathbf{0.8271 \pm 0.0115}$	0.7990 ± 0.0218	$\mathbf{0.5040 \pm 0.0194}$	$\mathbf{0.3685 \pm 0.0200}$	$\mathbf{0.8305 \pm 0.0135}$	$\mathbf{0.9017 \pm 0.0093}$

	Low-Medium					
Model	**ACC**	**RC**	**F1S**	**PR**	**SP**	**AUC**
LR	0.7597 ± 0.0064	0.7993 ± 0.0086	0.7668 ± 0.0063	0.7369 ± 0.0074	0.7209 ± 0.0100	0.8391 ± 0.0052
SVM	0.8681 ± 0.0102	0.8734 ± 0.0120	0.8677 ± 0.0086	0.8621 ± 0.0147	0.8628 ± 0.0146	0.5196 ± 0.0075
Decision Tree	0.8382 ± 0.0061	0.8419 ± 0.0084	0.8372 ± 0.0065	0.8327 ± 0.0066	0.8346 ± 0.0068	0.8383 ± 0.0062
Random Forest	0.8828 ± 0.0043	$\mathbf{0.9097 \pm 0.0066}$	0.8847 ± 0.0043	0.8611 ± 0.0064	0.8565 ± 0.0075	0.9587 ± 0.0022
GB	0.7746 ± 0.0054	0.8162 ± 0.0082	0.7816 ± 0.0054	0.7499 ± 0.0060	0.7339 ± 0.0078	0.8567 ± 0.0045
XGBoost	0.8329 ± 0.0057	0.8690 ± 0.0075	0.8372 ± 0.0055	0.8076 ± 0.0072	0.7976 ± 0.0089	0.9068 ± 0.0042
MLP	0.8615 ± 0.0057	0.8657 ± 0.0101	0.8607 ± 0.0061	0.8559 ± 0.0076	0.8574 ± 0.0085	0.9139 ± 0.0046
LightGBM	0.8175 ± 0.0050	0.8521 ± 0.0074	0.8220 ± 0.0050	0.7939 ± 0.0060	0.7838 ± 0.0074	0.8946 ± 0.0042
Meta-Model: LR	$\mathbf{0.8858 \pm 0.0042}$	0.8984 ± 0.0058	$\mathbf{0.8863 \pm 0.0045}$	$\mathbf{0.8747 \pm 0.0052}$	$\mathbf{0.8734 \pm 0.0057}$	$\mathbf{0.9608 \pm 0.0012}$

8) I DO NOT ATTEND EXTRA MATHEMATICS CLASSES (ST297Q09TA)

The variable, 'REPEAT', having a negative correlation with performance, is somehow patent among students and hence is distinct from the variable ST297Q09TA (No_Extra_Math_Classes), which has a value of 0, which means a student does attend extra Mathematics classes has a complex effect. The high achievers who are seemingly deprived of any such additional classes (blue shades) may be able to argue that they are optimal in the subject and thus do not need any extra help in that area.

The analysis hints that the relation between these variables and the performance is more complex than simple cause-and-effect logic would suggest. High achievers seem to take advantage of the high availability of digital resources and regular pre-class study practices. Still, there appears to be no need for enough extra Mathematics classes. Also, the student's gender has some effect on predicting performance, with boys being more significant than others in anticipating excellence in Mathematics found in this dataset.

Low and Medium Performance Study in Mathematics: By interpreting the SHAP values for students categorized within the Low-Medium performance levels in Mathematics, we gain insights into the determinants of their academic achievements, as visualized in Figure 5. The SHAP value intensity, in Figures 5 (a) and 5 (c) (spanning from blue to red) reveals the predictive power of each variable, with red representing a more significant positive influence and blue a lesser or adverse one.

9) DIGITAL DEVICES WITH SCREENS AT HOME (ST253Q01JA0_7)

A student who has 6–10 devices with school encoded value '7' fits in the Medium category of Mathematics performance, and this is reflected in the SHAP value swarm plot (Figure 5 (a)). This finding indicates that having many devices might help students learn as long as the number is not excessive and enough technology is offered for the students' learning activities.

10) I DO GO TO EXTRA MATHEMATICS CLASSES (ST297Q09JA0_0)

The value '0' means that such people took extra Mathematics lessons. Regarding heat map context (Figure 5 (c)), in this

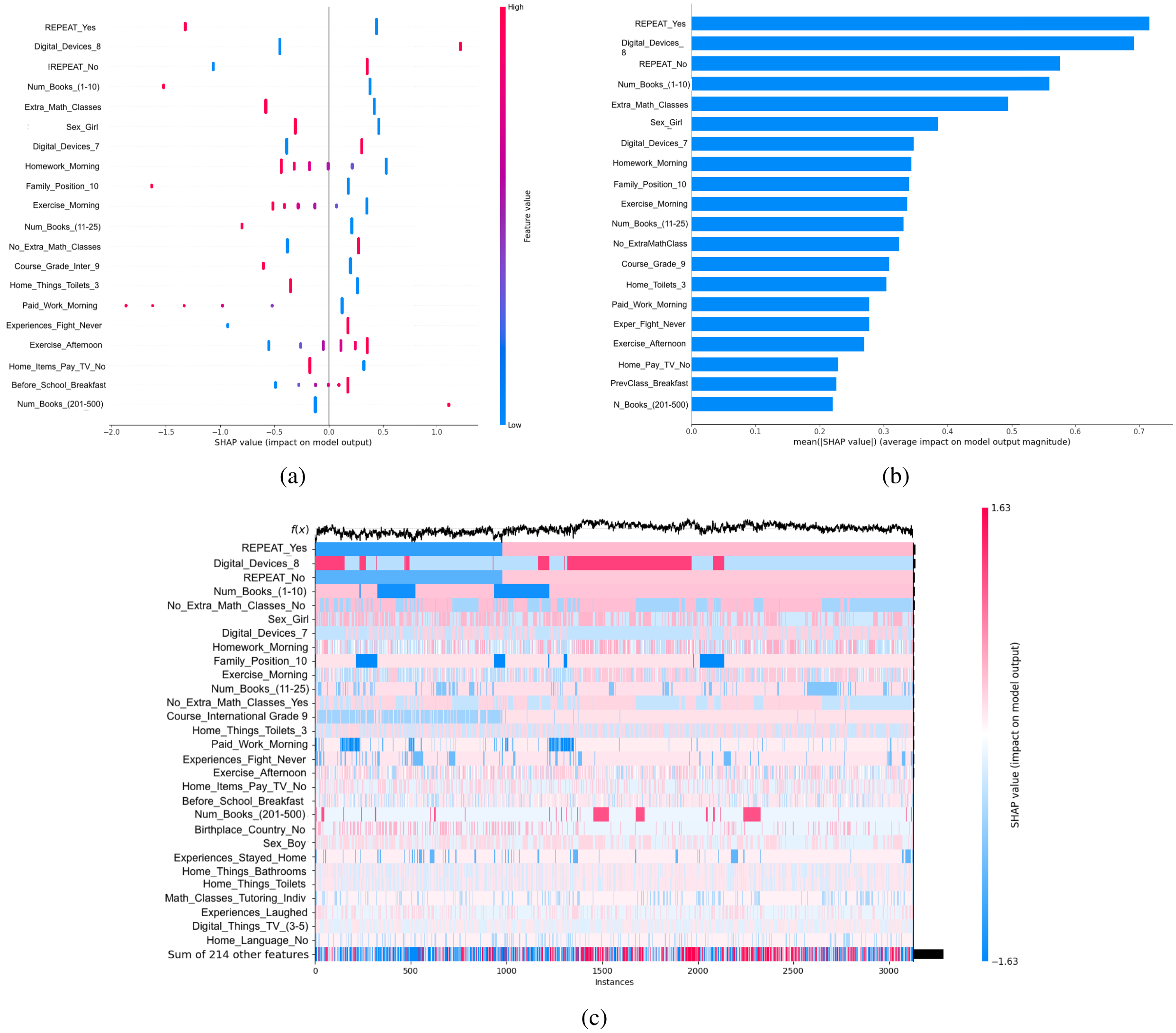


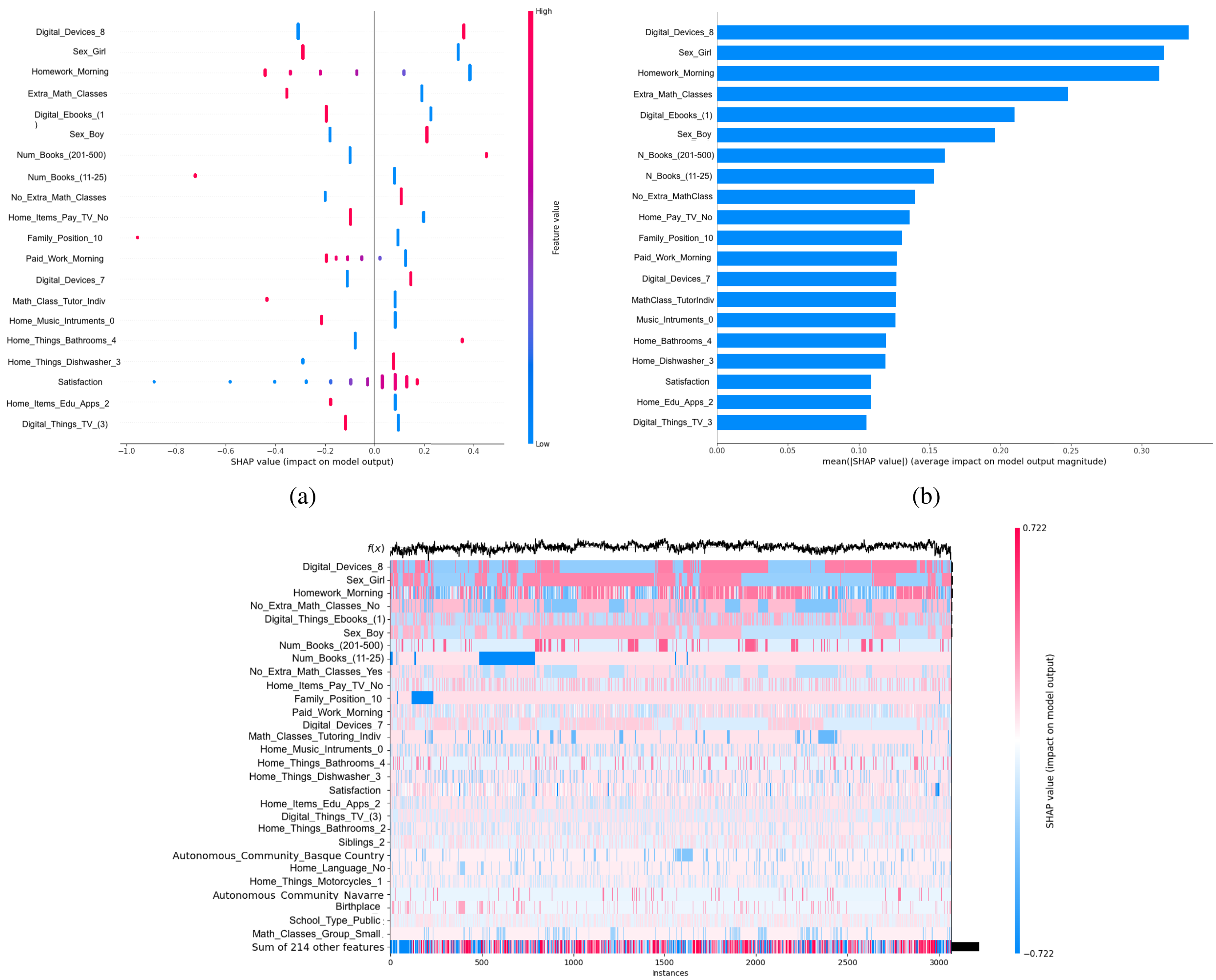
FIGURE 3. SHAP value distributions by attributing them to student performance affecting features. (a) The swarm plot depicts the dispersion and effect of each attribute for students' regime journalism (Low-performing students), where the x-axis shows the effect and the orientation of the attribute effect. (b) Column graph depicting the mean effect of the leading features for high-performing students with the graph bars representing the degree of contribution of the feature. (c) Heatmap of SHAP values for each feature and several sample sets, with the intensity of red (the positive impact on student performance), the intensity of blue, low impact.

particular variable, if red, it implies that taking those factors positively decreases performance in this Low-Medium area or borderline class.

B. MODEL EXPLAINABILITY

The performance of the predictive models or two groups of students is shown in Figure 6: Low-High (a), Medium-High (b), and Low-Medium (c) achievement levels. Inflection marks on the curves, encircled by vertical lines, indicate feature counts where model ACC improved relative to the after this feature count, depicting these features take a central idea in the model explainability. Thus, the SHAP values ranked the features in interference to their importance, and deletion of individual features continued to see their influence over time, culminating in the curves depicted. In the case of Low-High performance students, 20 features are the point at which there is an optimum feature cutoff, above which any additional features result in a drop in both ACC and AUC metric (in Figure 6 (a)). The relatively Medium level students perform better using up to 35 features, in Figure 6 (b), while Low and Medium level performance students cutoff has also been found at 20 (in Figure 6(c)). This selection is done so that there is a minimum of features and a maximum of predictive power, choosing the features that explain the students' levels.

The following research described in the Table 4, deals with how weight or importance of the features is calculated in the model rather than what is defined using graphics at this level. It indicates that a limited range of variables can provide a far greater fraction of the model's predictive power. Such results are beneficial when embedding such variables in a dashboard application.



(c)

FIGURE 4. **SHAP value visualization for features influencing Medium to High student performance in Mathematics. (a) The swarm plot depicting the centroids as descriptive statistics also tells pictorially about the coordinate-wise statistical influence the number of features has on the students demonstrating Medium High-performance average, and the two pole range with a degree on the x-axis indicates the range and polarity of the influence of each feature. (b) A bar chart shows the average SHAP score for features most often found in students with moderate to High scores. The lengths of the bars show how important certain feature influences are about the others. (c) Heatmap exhibiting the distribution of SHAP features across many observations, with each feature having distinct color-coded individual SHAP values with a fashion gradient from blue color on the left to red color on the right showing the descriptive ordering from least contribution to performance to highest contribution to performance concerning student performance factors.**

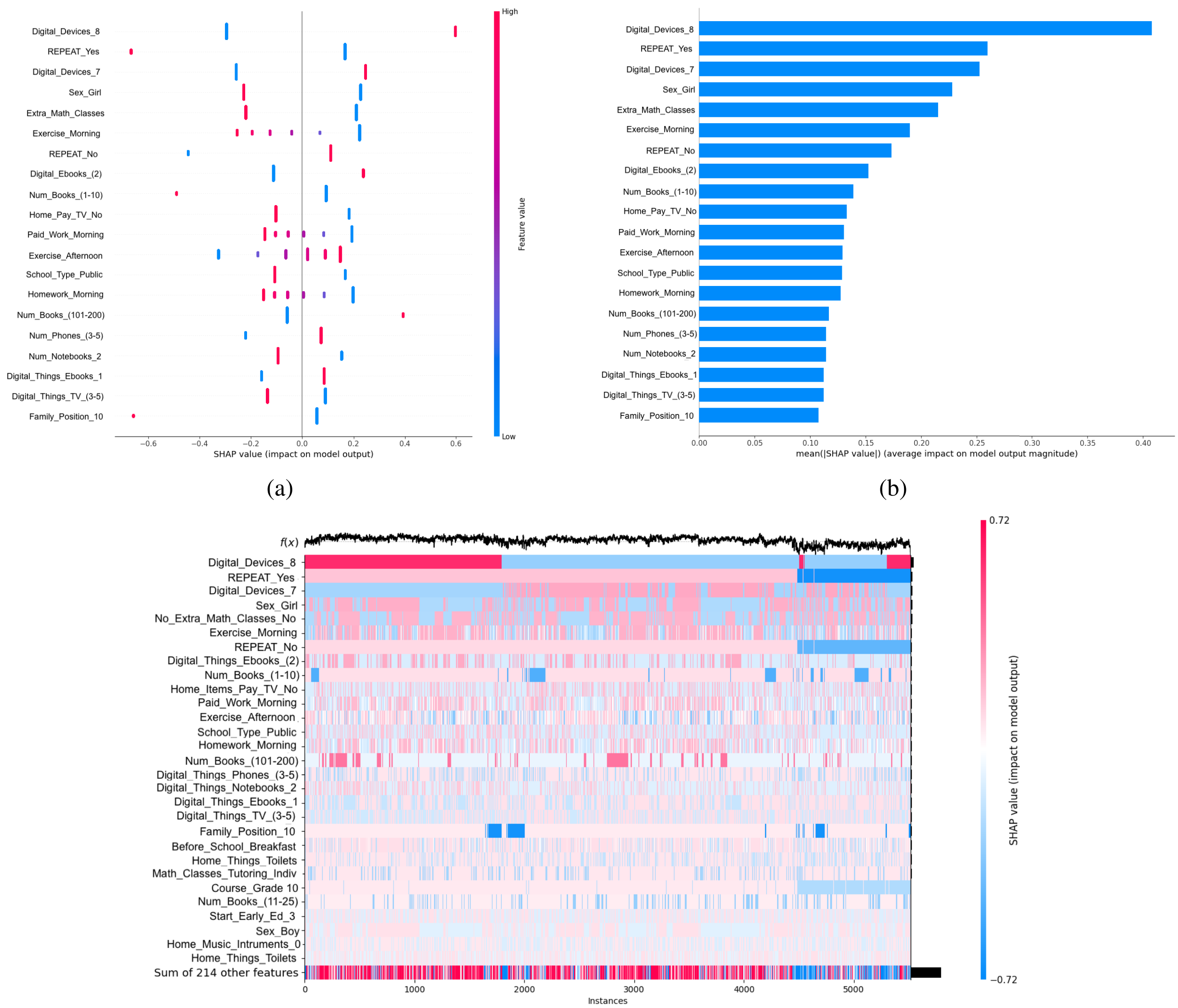
The application would be improved by considering these factors for further investigation and better insight into the trends and factors affecting student performance. This feature subset selection is essential not only for reducing the complexity of the model but also for maintaining the ACC of predictions. The model would be optimally constructed using the most critical variables, as identified during the SHAP analysis. This would render the model suitable for an interactive dashboard with real-time analysis using and interpreting data that embodies the highlights of the dataset.

C. APPLICATION OF PREDICTIVE DASHBOARD

The analysis in Figure 6 sought to establish whether the causes that identify variation in student performance could be simplified into a few main factors. This selection of educational variables or factors simplifies the number of variables, thus making the dashboard design more practical. Concentrating the number of primary variables in the dashboard application makes interacting with these educational factors possible, which could help educators.

The dashboard is focused on the identified key features and, therefore, provides a window into how student performance has changed over time and how it is expected to change in the future. This means that the dashboard mainly focuses on the communication of essential metrics, predicting scores, and the profiles of students so that effective and efficient decisions can be made on those students.

Within the broader interpretation of Figure 7 (a), the core characteristics of the dashboard aid in the decision-making processes. For example, the points 1-10 features, which are detailed as the top ten features, could be all widgets in the dashboard. Each widget will provide an independent piece of



(c)

FIGURE 5. Visualizes the SHAP value concerning the features affecting Low to Medium student performance in Mathematics. (a) The Swarm plot presents the variability and effect of the features on low intermediate-level students as shown on the x-axis, which presents the magnitude and direction of the feature impact. (b) Bar chart showing the average SHAP value for the most important features for final year undergraduate students who rank Low to Medium performance level with the lengths of the bars of every feature and the relative impact of that feature. (c) Heatmap showing the individual feature SHAP values over several instances along with an external matrix heatmap indicating the impact of each feature of the model on student performance via a color scale from blue to red, moving from Low impact to High impact, respectively.

information as follows: (1) **DR Projection**, this component presents UMAP in three Uniform Manifold dimensions representing the test dataset. Each point on the scatter plot corresponds to one student and is color-coded to show student population clusters. (2) **SHAP Values Bar Chart**, this is a bar chart that shows the SHAP values of the student's data (by selecting a point in the UMAP projection that corresponds to that student), which tells the relative importance of features in a model specifically for this student's data. (3) **Repeat Status Distribution**, this panel presents a basic bar graph that explains what proportion of students have repeated a grade (Repeat) and studied without repetition (No Repeat), thereby shedding some light on how widespread grade repetition is. (4) **Student Level Gauge**, which is also presented in a speedometer gauge, this visual depicts the level distribution of students in three achievement levels: Low (red), Medium (yellow), and High (green), to provide a broad overview of attainment levels of students. (5) **Gender Distribution Pie Chart**, this pie chart is designed to look at the gender proportions of the student population in that study. (6) **School Type Histogram**, it uses a histographic representation of students in the database who attended either public or private schools to explain the type of education provided. (7) **Autonomous Community Map**, the map feature gives geographical context by pinpointing the location of the autonomous community or communities where the students are from, enhancing the demographic analysis with spatial insights. (8) **Extra Mathematics Classes Pie Chart**, a pie chart indicating the percentage of students who do or do not attend extra Mathematics classes, which may correlate

TABLE 4. The evaluation of the SHAP values for the main features across three comparing schemes: Low-High, Medium-High, and Low-Medium. These SHAP values are reflected in how the features were used in the model. This shows the contribution of each feature within the model for prediction, where higher SHAP values are expected to drive the model more.

Low-High		Medium-High		Low-Medium	
Feature Name	SHAP Value	Feature Name	SHAP Value	Feature Name	SHAP Value
REPEAT_Yes	0.715068	Digital_Devices_8	0.333008	Digital_Devices_8	0.407681
Digital_Devices_8	0.690650	Sex_Girl	0.315830	REPEAT_Yes	0.259651
REPEAT_No	0.575782	Before_School_Homework_Morning	0.312381	Digital_Devices_7	0.252260
Num_Books_(1-10)	0.559799	No_Extra_Math_Classes_No	0.247907	Sex_Girl	0.227813
No_Extra_Math_Classes_No	0.494481	Digital_Things_Ebooks_1	0.209814	No_Extra_Math_Classes_No	0.214909
Sex_Girl	0.383587	Sex_Boy	0.196258	Before_School_Exercise_Morning	0.189449
Digital_Devices_7	0.346426	Num_Books_(201-500)	0.160643	REPEAT_No	0.173104
Before_School_Homework_Morning	0.342488	Num_Books_(11-25)	0.152889	Digital_Things_Ebooks_2	0.152284
Family_Position_10	0.339724	No_Extra_Math_Classes_Yes	0.139609	Num_Books_(1-10)	0.138859
Before_School_Exercise_Morning	0.337055	Home_Items_Pay_TV_No	0.136042	Home_Items_Pay_TV_No	0.132681
Num_Books_(11-25)	0.331356	Family_Position_10	0.130577	Before_School_Work_Morning	0.130393
No_Extra_Math_Classes_Yes	0.324851	Before_School_Work_Morning	0.126790	After_School_Exercise_Afternoon	0.128860
Course_(International Grade 9)	0.307893	Digital_Devices_7	0.126456	PRIVATESCH_public	0.128581
Home_Things_Toilets_3	0.304841	Math_Classes_Tutoring_Indiv_Yes	0.126084	Before_School_Homework_Morning	0.127504
Before_School_Paid_Work_Morning	0.277130	Home_Things_Musical_Instruments_1	0.125979	Num_Books_(101-200)	0.116626
Experiences_Fight_Never	0.276100	Home_Things_Bathrooms_4	0.119082	Digital_Things_Cell_Phones_3	0.114324
After_School_Exercise_Afternoon	0.268746	Home_Things_Dishwasher_3	0.118799	Digital_Devices_Notebooks_2	0.114109
Home_Items_Pay_TV_No	0.228096	Satisfaction	0.108655	Digital_Things_Ebooks_1	0.112193
Before_School_Breakfast	0.226407	Home_Items_Edu_Apps_No	0.108583	Digital_Things_TV_3	0.112041
Num_Books_(201-500)	0.220608	Digital_Things_TV_3	0.105437	Family_Position_10	0.107453

with academic performance in that subject. (9) **Digital Device Ownership Bar Chart**, this histogram categorizes students by the number of digital devices with screens they have access to at home, with categories ranging from 0 to more than 10, including recognizing the potential impact of digital access on educational opportunities. (10) **Home Book Quantity Histogram**, this bar chart quantifies the number of books students have in their homes, then sorted into several ranges from 0 to more than 500, acknowledging the role of educational resources available at home in academic performance.

Figure 7 (b) shows the usability of the dashboard where the user is shown how to filter a given area using the example of the autonomous community of Basque Country. Whenever a user clicks on the red region show (in this example containing the name Basque Country), all the Dashboard areas become relevant because specific changes are carried out based on the information stored in the panels pertinent to the selected region. In this case, all the information concerning the autonomous community of the Basque Country is shown, both at the level of data analysis and at the level of visualization of the latent space of the students of this autonomous community. In addition, in this latent space it is possible to select and analyze the variables that classify this student in this educational level in Mathematics.

This makes it possible for stakeholders to investigate education statistics in an advanced way, focusing on the information they want to promote and some of the issues in a particular community. By allowing such a narrow focus on one region, the dashboard becomes a means of enhancing understanding and decision-making.

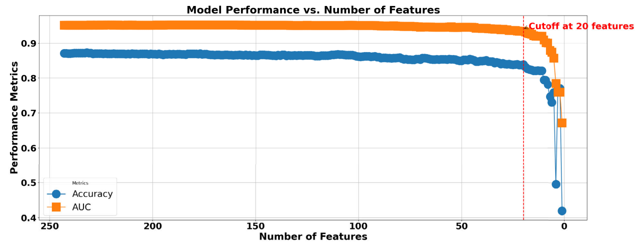
Figure 7 (c) illustrates the analytical capability of the application system. Interaction with the 3D UMAP projection by selecting an enrolled student leads to the presentation of the SHAP values in detail. The SHAP feature determines and quantifies the standard variables that are important for predicting the model used on a particular enrolled student.

After selecting a data point representing a student and clicking on it, the student profile dashboard is displayed. After that, the topological positioning of the selected student attributes, such as autonomous community, gender, school type, number of digital devices, and extra Mathematics class attendance, is displayed.

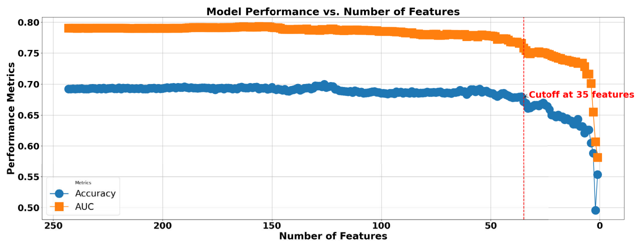
The dashboard depicts a bar chart in which each variable is represented by a bar whose length represents the magnitude of the SHAP value, and the color indicates the direction of the influence. Variables whose SHAP values are positive (red) are expected to improve performance, while those with negative values (blue) are expected to lower performance.

V. DISCUSSION, LIMITATIONS AND FUTURE WORKS

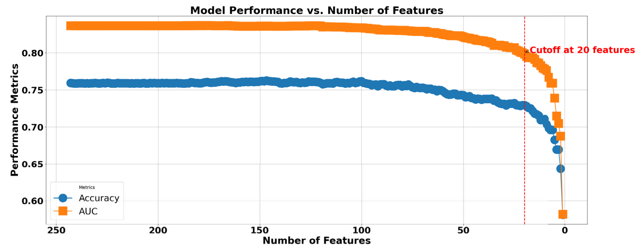
As derived from our detailed examination of the PISA 2022 dataset for Spain, some valuable factors of student performance in Mathematics were identified. Such tasks have been tackled with the application of eight binary classification models plus an ensemble meta-model, allowing a constrained interpretation of the hierarchal relation that exists among the host of educational, demographic, and resource provision factors. Nevertheless, metamodels continued to demonstrate their real power relative to isolated models and thus provided further support for the continued use of hierarchical techniques in educational data mining. In particular, the analysis of variables aided by the SHAP value analysis managed to classify specific prominent variables like course repeat student, owning a digital device, and extra out-of-class Mathematics lessons relevant to the comprehension of mathematical skills. There are great expectations for this activity, particularly for students who are repeaters by their inherent constitution or students who lack the means to enhance the content of the education they receive.



(a)



(b)



(c)

FIGURE 6. Analysis performance associated with the number of features is also carried out in a comparative manner considering different levels of student performance. The features retain a rank order from the most winning SHAP values to the least winning. Next, they were removed, step-down, beginning with the least disturbing ones. Plot (a) depicts Low-High-performance students who notice a significant drop in both ACC and AUC metrics at a cut-off of 20 features. Plot (b) focuses on the Medium-High performance students who experienced a relative decrease in performance when the cut-off for the features was at 35. Finally, panel (c) demonstrates Low-Medium performance students and has shown the same trend of reduction in performance at a cut-off of 20 features.

When contrasted with previous research, we found that our results corroborate the attention that literature gives to socioeconomic standing and the quality of teaching as the most important predictors of Mathematics performance [21], [22]. However, our work uniquely integrates these factors with technology, indicating that a child's digital device ownership tends to enhance his or her mathematical skills if there are effective teaching practices in place. This further corroborates existing studies [24], [31], which emphasize the potentiality of educational technologies but at a cost. In contrast, our study adds to the literature base as it brings out new findings based on machine learning.

Our research points out how home educational resources have an impact, as has been shown by [36] and [39], but adds a new dimension by allowing the interplay of factors such as out-of-class lessons and various demographic characteristics. Further, whereas gender gaps in Mathematics have been amply noted [34], [35], our research goes by segmenting performance, limiting the achievement gaps to a Low, Medium, and High modal using varying factors determining performance of the subjects on data-based approach allowing for a more nuanced understanding of achievement gaps.

It becomes clear that the models have not only become apparent for too many injustices in the negative and positive characteristics detached from higher levels of education but have also classified all influence factors into Low, Medium, and High extremes of performance. The apparent reason is that this is insufficient to create an all-encompassing policy on educational issues. As for those who were spared from generalizing and did accept variations, our pursuit of this study also invites the possibility of further studies investigating gender and study habit influences on attainment to address how to enhance students' academic achievement.

Limitations: This study is not without limitations. Firstly, the analysis is based solely on the PISA 2022 dataset, which, while offering the latest insights, does not allow for direct longitudinal comparisons with previous years. Incorporating datasets from earlier assessments, such as PISA 2018, would enable a broader perspective on trends and changes over time. However, this was not feasible in the present study due to variable differences and the need to focus on new variables introduced in 2022. Secondly, the study relies on SHAP values for interpretability, which, while powerful, may not capture all potential causal relationships within the dataset. Future work could explore alternative interpretability techniques or experimental validation to strengthen the findings. Lastly, while the developed dashboard offers significant utility, its effectiveness has not yet been tested in real-world educational settings. Implementation and usability testing with stakeholders such as policymakers and educators remain necessary to fully validate its impact.

The ability of the model to provide an adequate prediction to the meta-model and the functional usability of our dashboard are all just the beginning of the studies in this field. The future evaluation will go beyond the one reported in this paper and employ time series analysis to monitor the distribution of performance changes among students within the same cohort over time using, ideally, a more extensive range of variables. Also, more aspects of attention given to the relationship causation and potential evidence explaining them might be returned, especially suggesting an experimental approach or qualitative-quantitative mix.

The future work will evaluate other methods for class imbalance, including oversampling and mixed techniques, to determine if any of these methods are better than undersampling.

Regarding prospects, the dashboard usability must also include the analysis of uplifts on forecast or situation planning, which, perhaps, the actual strength lies. Such advancement would enable educationists to project the expected impacts of custom-made strategies, resulting in more efficient student cultivation. In addition, incorporating expansive and more complicated educational datasets to utilize more advanced models and more features will further contribute to disaggregated findings that would promote better and specific education for varied learning settings.

FIGURE 7. Real-time predictive dashboard. (a) Interactive toolkit focused on educational management solutions. (b) A system for filtering out data for the case studies based on geographical location. (c) A widget that describes a particular student and the most important attributes.

TABLE 5. PISA mathematics survey variables (Part 1). This table catalogs the first part of the variables collected in the PISA. When students of maths were surveyed, information concerning their home surroundings and digital technologies was collected in the first two questionnaires. It covers items from items possessed in households and scenarios of communication through the internet, all explained with data type.

Variable	Short name	Description	Data types
ST001D01T	Course	What course are you studying?	Categorical
ST004D01T	Sex	Sex of student, answer with male and female	Binary
ST250	Home_Items	Do you have the following items in your home? Answer yes or no	
ST250Q01TA	Room_Alone	A room for you alone	Binary
ST250Q02TA	Computer	A computer that you can use to study	Binary
ST250Q03TA	Edu_Apps	Educational computer applications or programs	Binary
ST250Q04TA	Cell_Phone	A cell phone you own with Internet access	Binary
ST250Q05TA	Internet	Access to Internet	Binary
ST250Q06TA	Movie_Platforms	Series and movie platforms (HBO, Netflix)	Binary
ST250Q07TA	Pay_TV	Pay TV	Binary
ST251	Home_Things	How many of the following things are in your home? Answer with 0, 1, 2, and 3 or more	
ST251Q01TA	Cars	Cars, vans or trucks	Categorical
ST251Q02TA	Motorcycles	Mopeds or motorcycles	Categorical
ST251Q03TA	Bathrooms	Bathrooms with bathtub or shower	Categorical
ST251Q04TA	Toilets	Toilets	Categorical
ST251Q06TA	Musical_Instruments	Musical instruments	Categorical
ST251Q07TA	Art	Works of art	Categorical
ST251Q08TA	Dishwasher	Dishwasher	Categorical
ST251Q09TA	Parking	Parking spaces	Categorical
ST253Q01JA0	Digital_Devices	How many digital devices with screens are there in your home? Answer with 0, 1, 2, 3, 4, 5, 6-10 and more than 10	Categorical
ST254	Digital_Things	How many of the following digital devices are in your home? Answer with 0, 1-2, 3-5, more than 5, and I do not know	
ST254Q01TA	TV	TV	Categorical
ST254Q02TA	Desktop_Computers	Desktop computers	Categorical
ST254Q03TA	Notebooks	Notebooks	Categorical
ST254Q04TA	Tablets	Tablets	Categorical
ST254Q05TA	Ebooks	E-books	Categorical
ST254Q06TA	Cell_Phones	Cell phones with Internet access	Categorical
ST255Q01JA	Num_Books	How many books are in your house? Answer with 0, 1-10, 11-25, 26-100, 101-200, 201-500, and more than 500 books	Categorical
ST230Q01JA0	Siblings	How many siblings do you have? Answer with 0, 1, 2, and 3 or more	Categorical
ST259Q01JA0	Family_Position	In what position would you place your family at this time? Answer from 1-10	Scalar
ST019Q01JA0	Birthplace	In which community or country were you and your parents born?	Categorical
ST022Q01JA0	Home_Language	What language do you speak at home most of the time?	Categorical
ST226Q01JA0	Time_In_Center	How long have you been with this center? Answer with 3 or more, 2, 1, at the beginning of this course, and once this course has started	Categorical
ST125Q01JA0	Start_Early_Ed	How old were you when you started Early Childhood Education?	Categorical
ST062	Class_Attendance	In the last two full weeks of class, how often did the following things happen? Answer with 0, 1-2, 3-4, and more than 4	
ST062Q01JA0	Skipped_Day	I skipped a whole day	Categorical
ST062Q02JA0	Skipped_Classes	I skipped some classes	Categorical
ST062Q03JA0	Late	I arrived late to the center	Categorical

Future work should deepen studies to promote a more elaborate and flexible educational system that not only addresses the existing needs of students but can also predict their needs so that each of the different academic levels can reach their academic achievements.

VI. CONCLUSION

In conclusion, this study has not only contributed to the body of knowledge concerning educational Data Science. Still, it has also highlighted the power of ML models in dissecting complex datasets like PISA. With the help of modern analytics and interpretability methods like SHAP values, we have successfully extracted the significant determinants of students' performance in the proper context. In light of these findings, the developed interactive dashboard is offered as a resource for educators and policymakers, improving the quality of education through the effective utilization of data.

The information in this dashboard enables policymakers to understand and deal with regional inequality in student achievement by filtering results by location, gender, or other factors. It can also help to make well-informed decisions as far as the funding is concerned, for instance, directing resources to backward communities, formulating programs targeting specific types of underutilized students, and later evaluating the performance of the strategies over a period. By embedding these capabilities, the tool can facilitate

TABLE 6. PISA mathematics survey variables (Part II). Continuation of the variable catalog from Table 5, focusing more on powers of autonomous connectedness, as related to school and out-of-school activities of the respondent and their socio-demographic background. In this section, quantifiable creative items and yes/no responses applicable to the investigation are brought in.

Variable	Short name	Description	Data types
ST038	Experiences	During the past 12 months, how often have you had the following experiences at the center? Answer with never or almost never, several times a year, several times a month, and one or more times per week	
ST038Q03JA0	Exclusion	Other students have purposely excluded me	Categorical
ST038Q04JA0	Laughed	Other students have laughed at me	Categorical
ST038Q05JA0	Threatened	Other students have threatened me	Categorical
ST038Q06JA0	Taken_Things	Other students have taken or broken my things	Categorical
ST038Q07JA0	Hit	Other students have hit or pushed me	Categorical
ST038Q08JA0	Rumors	Other students have spread horrible rumors about me	Categorical
ST038Q09JA0	Fight	I was in a fight inside the center	Categorical
ST038Q10JA0	Stayed_Home	I stayed at home instead of going to class because I did not feel safe	Categorical
ST038Q11JA0	Gave_Money	I gave money to someone at the center because they threatened me	Categorical
ST294	Before_School	In a typical school week, how often do you do the following before class? Answer with 0, 1, 2, 3, 4, and 5 or more	
ST294Q01JA0	Breakfast	Breakfast	Categorical
ST294Q02JA0	Homework_Morning	Studying or doing homework	Categorical
ST294Q03JA0	Work_Morning	Working in the home or caring for family members	Categorical
ST294Q04JA0	Paid_Work_Morning	Work in exchange for money.	Categorical
ST294Q05JA0	Exercise_Morning	Exercising or playing a sport	Categorical
ST295	After_School	In a typical school week, how many days do you do the following when you leave the center? Answer with 0, 1, 2, 3, 4, and 5 or more	
ST295Q01JA0	Dinner	Dinner	Categorical
ST295Q02JA0	Homework_Afternoon	Studying or doing homework	Categorical
ST295Q03JA0	Work_Afternoon	Working in the home or caring for family members	Categorical
ST295Q04JA0	Paid_Work_Afternoon	Work in exchange for money.	Categorical
ST295Q05JA0	Exercise_Afternoon	Exercising or playing a sport	Categorical
ST016Q01JA0	Satisfaction	to what extent are you satisfied these days with your whole life? Answer from 1 to 10	Scalar
ST297	Math_Classes	This year, what types of extra Mathematics classes are you participating in? Answer yes or no	
ST297Q01TA	Tutoring_Indiv	Individual tutoring with one person	Binary
ST297Q03TA	Tutoring_Online	Internet or computer-based tutoring through a program or application	Binary
ST297Q05TA	Video_Classes	Videotaped classes taught by one person	Binary
ST297Q06TA	Group_Small	Small group study or practice (2 to 7 students)	Binary
ST297Q07TA	Group_Large	Large group study or practice (8 or more students)	Binary
ST297Q09TA	No_Extra_Math_Classes	I do not go to extra Mathematics classes	Binary
LANGN	Language	Language of the questionnaire. Answer with Spanish, Basque, Catalan, Valencian, Galician, Aranes, and Another language	Categorical
REPEAT	Repeater	Repeating student. Answer with yes or no	Binary
PRIVATESCH	School_Type	Type of center. Answer with public or private	Binary
Mathematics scores	Math_Score	Mathematics score prediction label with 3 categories (Low, Medium, and High)	Categorical

evidence-based decision-making, enhancing fairness and efficiency in the education system.

APPENDIX

A. PISA SURVEY VARIABLES

A considerable set of variables has been introduced in this PISA survey concerning students' mathematical literacy to collect sufficient data on the factors that influence the student's ability to perform. Table 5 shows the students' household settings and their computer skills variables. These variables cover a variety of data types, which include the binary presence or absence of certain items (for ST250Q01TA, the item signifies if a spare room is designated for study), whereas for some, fussy categorical variables revolve around a list of responses (e.g., ST251 lists cars, number of rooms, among others.).

Moving forward in Table 6, the students were questioned about their biography, daily activities, and societies which included bullying at school (e.g., ST038Q03JA) and life contentment levels (ST016Q01JA). Such a wide range of data makes it possible to study different aspects that contribute to the final educational results.

The parameters tabulated in these tables are particularly significant in contextualizing the place and nature of students' learning. For instance, ownership of mobile phones or computers (ST253Q01JA0 in Table 5) can be an antecedent of the overall level of Internet skills. In contrast, the rate of specific negative social experiences such as these (ST038 in Table 6) can influence the degree of students' active participation and the nature of their well-being. Information of this kind is adequate when defining educational policies and interventions to establish favorable conditions for teaching and learning and meet their requirements.

TABLE 7. The detailed comparison of eight ML models, hyperparameter combinations, and a meta-model. This analytical period is based on ACC, RC, F1S, PR, SP, and AUC. Turning to the included results, they furnish some gradation in the models and their level of discrimination of students in terms of Low vs. High academic achievements. It is also worth mentioning that the meta-model characteristic, indicated with an asterisk, is an ensemble of all models trained to perform on various hyperparameters that are optimal for the given task. However, there is no standard deviation or bootstrap percentage for the mean and standard deviation scores for all metrics used in this study for model comparison.

Model	Hyperparameters	ACC	RC	F1S	PR	SP	AUC
LR	'penalty': 'l1', 'C': 1	0.8777 ± 0.0080	**0.9108 ± 0.0175**	0.6157 ± 0.0181	0.4653 ± 0.0193	0.8737 ± 0.0089	0.9575 ± 0.0044
	'penalty': 'l1', 'C': 10	0.8792 ± 0.0066	0.9058 ± 0.0170	0.6174 ± 0.0158	0.4685 ± 0.0175	0.8761 ± 0.0076	0.9551 ± 0.0047
	'penalty': 'l1', 'C': 100	0.8791 ± 0.0069	0.9087 ± 0.0163	0.6178 ± 0.0148	0.4683 ± 0.0165	0.8755 ± 0.0081	0.9551 ± 0.0047
	'penalty': 'l2', 'C': 1	0.8799 ± 0.0064	0.9096 ± 0.0170	**0.6197 ± 0.0151**	0.4702 ± 0.0166	0.8764 ± 0.0075	**0.9576 ± 0.0045**
	'penalty': 'l2', 'C': 10	**0.8802 ± 0.0071**	0.9036 ± 0.0175	0.6187 ± 0.0156	**0.4706 ± 0.0175**	**0.8774 ± 0.0082**	0.9557 ± 0.0042
	'penalty': 'l2', 'C': 100	0.8795 ± 0.0074	0.9034 ± 0.0172	0.6172 ± 0.0178	0.4690 ± 0.0193	0.8766 ± 0.0082	0.9551 ± 0.0050
SVM	'kernel': 'linear', 'C': 1	0.8756 ± 0.0079	0.9098 ± 0.0166	0.6114 ± 0.0171	0.4606 ± 0.0184	0.8715 ± 0.0088	0.9546 ± 0.0046
	'kernel': 'linear', 'C': 10	0.8759 ± 0.0073	0.9055 ± 0.0148	0.6108 ± 0.0164	0.4611 ± 0.0178	0.8723 ± 0.0083	0.9520 ± 0.0046
	'kernel': 'linear', 'C': 100	0.8753 ± 0.0074	0.9089 ± 0.0189	0.6105 ± 0.0173	0.4599 ± 0.0180	0.8713 ± 0.0082	0.9523 ± 0.0058
	'kernel': 'rbf,' 'C': 1	0.8828 ± 0.0072	0.9258 ± 0.0168	0.6295 ± 0.0159	0.4772 ± 0.0178	0.8776 ± 0.0084	0.9647 ± 0.0042
	'kernel': 'rbf', 'C': 10	**0.9005 ± 0.0062**	**0.9334 ± 0.0145**	**0.6686 ± 0.0166**	**0.5212 ± 0.0194**	**0.8966 ± 0.0069**	**0.9717 ± 0.0038**
	'kernel': 'rbf,' 'C': 100	0.9001 ± 0.0064	0.9327 ± 0.0151	0.6676 ± 0.0167	0.5201 ± 0.0187	0.8962 ± 0.0070	0.9714 ± 0.0047
Decision Tree	'criterion': 'entropy', 'max_depth': None	0.8344 ± 0.0094	0.8911 ± 0.0202	0.5365 ± 0.0173	0.3840 ± 0.0163	0.8276 ± 0.0102	0.8593 ± 0.0112
	'criterion': 'entropy', 'max_depth': 5	0.7787 ± 0.0360	0.8649 ± 0.0490	0.4597 ± 0.0327	0.3152 ± 0.0359	0.7684 ± 0.0451	**0.8868 ± 0.0093**
	'criterion': 'entropy', 'max_depth': 10	0.8084 ± 0.0140	0.8958 ± 0.0260	0.5017 ± 0.0173	0.3488 ± 0.0171	0.7979 ± 0.0172	0.8731 ± 0.0114
	'criterion': 'gini', 'max_depth': None	**0.8349 ± 0.0097**	0.8902 ± 0.0199	**0.5369 ± 0.0180**	**0.3846 ± 0.0172**	**0.8282 ± 0.0108**	0.8592 ± 0.0109
	'criterion': 'gini', 'max_depth': 5	0.7813 ± 0.0311	0.8627 ± 0.0436	0.4612 ± 0.0269	0.3163 ± 0.0298	0.7715 ± 0.0390	0.8848 ± 0.0081
	'criterion': 'gini', 'max_depth': 10	0.8129 ± 0.0118	**0.9023 ± 0.0210**	0.5093 ± 0.0173	0.3551 ± 0.0167	0.8022 ± 0.0140	0.8622 ± 0.0116
Random Forest	'n_estimators': 50, 'max_depth': None	0.8958 ± 0.0071	0.9276 ± 0.0173	0.6569 ± 0.0180	0.5089 ± 0.0211	0.8920 ± 0.0082	0.9722 ± 0.0045
	'n_estimators': 50, 'max_depth': 5	0.8129 ± 0.0114	0.9060 ± 0.0189	0.5103 ± 0.0167	0.3554 ± 0.0160	0.8017 ± 0.0132	0.9370 ± 0.0066
	'n_estimators': 100, 'max_depth': None	**0.8965 ± 0.0064**	**0.9315 ± 0.0127**	**0.6594 ± 0.0158**	**0.5106 ± 0.0180**	**0.8923 ± 0.0072**	0.9737 ± 0.0036
	'n_estimators': 100, 'max_depth': 5	0.8153 ± 0.0127	0.9088 ± 0.0176	0.5144 ± 0.0192	0.3590 ± 0.0182	0.8040 ± 0.0145	0.9383 ± 0.0067
	'n_estimators': 200, 'max_depth': None	0.8953 ± 0.0068	0.9313 ± 0.0172	0.6568 ± 0.0179	0.5075 ± 0.0198	0.8910 ± 0.0075	**0.9739 ± 0.0049**
	'n_estimators': 200, 'max_depth': 5	0.8150 ± 0.0093	0.9092 ± 0.0181	0.5138 ± 0.0139	0.3583 ± 0.0132	0.8036 ± 0.0109	0.9385 ± 0.0063
GB	'n_estimators': 100, 'learning_rate': 0.1	0.8744 ± 0.0079	0.9197 ± 0.0149	0.6116 ± 0.0175	0.4584 ± 0.0186	0.8689 ± 0.0087	0.9595 ± 0.0046
	'n_estimators': 100, 'learning_rate': 0.01	0.7992 ± 0.0165	0.8643 ± 0.0312	0.4812 ± 0.0175	0.3339 ± 0.0178	0.7914 ± 0.0207	0.9106 ± 0.0075
	'n_estimators': 200, 'learning_rate': 0.1	0.8871 ± 0.0059	0.9325 ± 0.0144	0.6397 ± 0.0147	0.4871 ± 0.0164	0.8816 ± 0.0067	0.9652 ± 0.0037
	'n_estimators': 200, 'learning_rate': 0.01	0.8226 ± 0.0110	0.8877 ± 0.0220	0.5185 ± 0.0155	0.3665 ± 0.0154	0.8148 ± 0.0134	0.9300 ± 0.0059
	'n_estimators': 300, 'learning_rate': 0.1	**0.8912 ± 0.0074**	**0.9370 ± 0.0135**	**0.6494 ± 0.0178**	**0.4972 ± 0.0197**	**0.8857 ± 0.0081**	**0.9670 ± 0.0039**
	'n_estimators': 300, 'learning_rate': 0.01	0.8389 ± 0.0088	0.8974 ± 0.0169	0.5450 ± 0.0163	0.3916 ± 0.0159	0.8318 ± 0.0098	0.9396 ± 0.0060
XGBoost	'n_estimators': 100, 'learning_rate': 0.1	0.8914 ± 0.0065	0.9375 ± 0.0137	0.6500 ± 0.0153	0.4977 ± 0.0174	0.8859 ± 0.0075	0.9678 ± 0.0037
	'n_estimators': 100, 'learning_rate': 0.01	0.8283 ± 0.0124	0.8999 ± 0.0218	0.5303 ± 0.0180	0.3763 ± 0.0190	0.8197 ± 0.0151	0.9325 ± 0.0062
	'n_estimators': 200, 'learning_rate': 0.1	0.8970 ± 0.0072	**0.9384 ± 0.0161**	0.6623 ± 0.0179	0.5120 ± 0.0205	0.8921 ± 0.0081	0.9705 ± 0.0044
	'n_estimators': 200, 'learning_rate': 0.01	0.8480 ± 0.0093	0.9114 ± 0.0161	0.5633 ± 0.0171	0.4079 ± 0.0176	0.8403 ± 0.0107	0.9458 ± 0.0046
	'n_estimators': 300, 'learning_rate': 0.1	**0.8984 ± 0.0071**	0.9366 ± 0.0148	**0.6647 ± 0.0184**	**0.5156 ± 0.0217**	**0.8938 ± 0.0081**	**0.9709 ± 0.0033**
	'n_estimators': 300, 'learning_rate': 0.01	0.8605 ± 0.0079	0.9242 ± 0.0160	0.5876 ± 0.0154	0.4309 ± 0.0160	0.8528 ± 0.0089	0.9544 ± 0.0047
MLP	**'hidden_layer_sizes': (100,), 'activation': 'tanh'**	**0.9019 ± 0.0077**	0.9332 ± 0.0150	**0.6719 ± 0.0193**	**0.5253 ± 0.0231**	**0.8982 ± 0.0088**	**0.9680 ± 0.0042**
	'hidden_layer_sizes': (100,), 'activation': 'relu'	0.8977 ± 0.0061	**0.9339 ± 0.0145**	0.6626 ± 0.0157	0.5137 ± 0.0179	0.8934 ± 0.0070	0.9670 ± 0.0042
	'hidden_layer_sizes': (100,), 'activation': 'logistic'	0.8805 ± 0.0103	0.9160 ± 0.0211	0.6228 ± 0.0206	0.4724 ± 0.0250	0.8762 ± 0.0127	0.9588 ± 0.0041
	'hidden_layer_sizes': (100, 50, 25), 'activation': 'tanh'	0.8971 ± 0.0075	0.9288 ± 0.0168	0.6601 ± 0.0197	0.5122 ± 0.0218	0.8933 ± 0.0080	0.9628 ± 0.0052
	'hidden_layer_sizes': (100, 50, 25), 'activation': 'relu'	0.8936 ± 0.0069	0.9298 ± 0.0157	0.6527 ± 0.0164	0.5031 ± 0.0192	0.8892 ± 0.0081	0.9645 ± 0.0040
	'hidden_layer_sizes': (100, 50, 25), 'activation': 'logistic'	0.8861 ± 0.0109	0.9246 ± 0.0200	0.6362 ± 0.0239	0.4856 ± 0.0280	0.8814 ± 0.0128	0.9551 ± 0.0059
LightGBM	'n_estimators': 50, 'learning_rate': 0.1	0.8873 ± 0.0068	0.9317 ± 0.0174	0.6401 ± 0.0161	0.4878 ± 0.0176	0.8820 ± 0.0076	0.9647 ± 0.0043
	'n_estimators': 50, 'learning_rate': 0.01	0.8305 ± 0.0111	0.8803 ± 0.0201	0.5278 ± 0.0196	0.3772 ± 0.0193	0.8245 ± 0.0127	0.9262 ± 0.0074
	'n_estimators': 100, 'learning_rate': 0.1	0.8968 ± 0.0068	**0.9394 ± 0.0146**	0.6619 ± 0.0165	0.5112 ± 0.0195	0.8916 ± 0.0080	0.9706 ± 0.0037
	'n_estimators': 100, 'learning_rate': 0.01	0.8431 ± 0.0104	0.8913 ± 0.0171	0.5501 ± 0.0197	0.3980 ± 0.0195	0.8373 ± 0.0115	0.9352 ± 0.0064
	'n_estimators': 200, 'learning_rate': 0.1	**0.8986 ± 0.0072**	0.9361 ± 0.0144	**0.6651 ± 0.0177**	**0.5162 ± 0.0211**	**0.8941 ± 0.0083**	**0.9725 ± 0.0041**
	'n_estimators': 200, 'learning_rate': 0.01	0.8589 ± 0.0090	0.9058 ± 0.0176	0.5800 ± 0.0180	0.4268 ± 0.0185	0.8532 ± 0.0101	0.9487 ± 0.0061
(Meta-Model: LR)*	'C': 0.1, 'solver':liblinear, 'penalty': 'l1'	0.9108 ± 0.0060	0.9378 ± 0.0147	0.6934 ± 0.0169	0.5503 ± 0.0200	0.9076 ± 0.0067	**0.9775 ± 0.0037**
	'C': 0.1, 'solver':liblinear, 'penalty': 'l2'	0.9049 ± 0.0059	**0.9407 ± 0.0138**	0.6802 ± 0.0162	0.5329 ± 0.0186	0.9006 ± 0.0065	0.9751 ± 0.0038
	'C': 0.1, 'solver':saga, 'penalty': 'l1'	0.9140 ± 0.0061	0.9364 ± 0.0139	0.7009 ± 0.0172	0.5603 ± 0.0208	0.9113 ± 0.0068	0.9772 ± 0.0036
	'C': 0.1, 'solver':saga, 'penalty': 'l2'	0.9108 ± 0.0057	0.9363 ± 0.0141	0.6930 ± 0.0157	0.5503 ± 0.0190	0.9077 ± 0.0066	0.9750 ± 0.0036
	'C': 1, 'solver':liblinear, 'penalty': 'l1'	0.9150 ± 0.0065	0.9330 ± 0.0140	0.7026 ± 0.0180	0.5638 ± 0.0226	0.9129 ± 0.0074	0.9771 ± 0.0034
	'C': 1, 'solver':liblinear, 'penalty': 'l2'	0.9133 ± 0.0061	0.9368 ± 0.0161	0.6992 ± 0.0176	0.5580 ± 0.0204	0.9105 ± 0.0066	0.9771 ± 0.0039
	'C': 1, 'solver':saga, 'penalty': 'l1'	0.9143 ± 0.0065	0.9314 ± 0.0132	0.7005 ± 0.0175	0.5618 ± 0.0222	0.9123 ± 0.0075	0.9767 ± 0.0032
	'C': 1, 'solver':saga, 'penalty': 'l2'	0.9147 ± 0.0076	0.9359 ± 0.0163	0.7026 ± 0.0216	0.5628 ± 0.0264	0.9122 ± 0.0084	0.9771 ± 0.0042
	'C': 10, 'solver':liblinear, 'penalty': 'l1'	0.9151 ± 0.0061	0.9323 ± 0.0160	0.7026 ± 0.0175	0.5640 ± 0.0211	0.9130 ± 0.0068	0.9769 ± 0.0039
	'C': 10, 'solver':liblinear, 'penalty': 'l2'	0.9140 ± 0.0067	0.9319 ± 0.0160	0.6998 ± 0.0180	0.5606 ± 0.0221	0.9118 ± 0.0075	0.9767 ± 0.0038
	'C': 10, 'solver':saga, 'penalty': 'l1'	0.9140 ± 0.0064	0.9328 ± 0.0157	0.7001 ± 0.0175	0.5607 ± 0.0217	0.9118 ± 0.0073	0.9768 ± 0.0041
	'C': 10, 'solver':saga, 'penalty': 'l2'	0.9152 ± 0.0062	0.9339 ± 0.0172	0.7032 ± 0.0177	0.5642 ± 0.0209	0.9130 ± 0.0067	0.9772 ± 0.0042
	'C': 100, 'solver':liblinear, 'penalty': 'l1'	0.9143 ± 0.0065	0.9346 ± 0.0138	0.7012 ± 0.0175	0.5615 ± 0.0220	0.9119 ± 0.0075	0.9770 ± 0.0035
	'C': 100, 'solver':liblinear, 'penalty': 'l2'	0.9151 ± 0.0059	0.9299 ± 0.0170	0.7019 ± 0.0172	0.5641 ± 0.0217	0.9133 ± 0.0070	0.9766 ± 0.0042
	'C': 100, 'solver':saga, 'penalty': 'l1'	**0.9157 ± 0.0066**	0.9330 ± 0.0158	**0.7042 ± 0.0184**	**0.5659 ± 0.0229**	**0.9136 ± 0.0076**	0.9766 ± 0.0040
	'C': 100, 'solver':saga, 'penalty': 'l2'	0.9153 ± 0.0076	0.9313 ± 0.0155	0.7029 ± 0.0211	0.5650 ± 0.0263	0.9134 ± 0.0085	0.9764 ± 0.0039

*Note: The meta-model presented in this table is constructed by integrating the most effective models, each optimized with their respective best hyperparameters.

B. LOW AND HIGH-PERFORMANCE STUDY

The results of this comparison are highlighted in Table 7 and include the additional evaluation for the meta-model, which does accept correspondence marked by an asterisk * for comments purposes only. This meta-model is an individual model that combines all the best-fine-tuned models for the attempted task. These outcomes represent the successful implementation of an ensemble meta-modeling strategy using all explanatory variables that are genuinely better than the best performance score. More specifically, it means the meta-model can improve on the AUC, which will help students differentiate classes without equal attention to performance measures. For educational organizations intending to employ predictive analytics to mitigate students exhibiting Low academic achievement in Mathematics, such enhancement could be of great help.

Superiority of the SVM with RBF Kernel. The SVM model, particularly concerning the use of the *kernel = 'rbf'*

TABLE 8. Analysis of eight ML models' comparison regarding hyperparameter settings about the meta-model but touches Medium to High academic performance range. Meta-model, or statistical synthesis, is marked with an asterisk. It follows Bussey and various other models and shifts away from the focus of each model that has been painstakingly set to its misconceptions for the present comparison. Also, Where PC was measured 100 times using Bootstrap Iteration, and the mean and standard deviation for all the metrics were evaluated for the more accurate performance of the models.

Model	Hyperparameters	ACC	RC	F1S	PR	SP	AUC
LR	**'penalty': 'l1', 'C': 1**	**0.7057 ± 0.0107**	**0.7489 ± 0.0263**	**0.3586 ± 0.0143**	**0.2358 ± 0.0106**	**0.7003 ± 0.0114**	**0.7956 ± 0.0129**
	'penalty': 'l1', 'C': 10	0.7038 ± 0.0105	0.7420 ± 0.0247	0.3549 ± 0.0131	0.2333 ± 0.0098	0.6991 ± 0.0117	0.7906 ± 0.0119
	'penalty': 'l1', 'C': 100	0.7017 ± 0.0102	0.7442 ± 0.0232	0.3540 ± 0.0118	0.2323 ± 0.0090	0.6964 ± 0.0116	0.7898 ± 0.0119
	'penalty': 'l2', 'C': 1	0.7040 ± 0.0093	0.7441 ± 0.0246	0.3558 ± 0.0125	0.2338 ± 0.0093	0.6991 ± 0.0104	0.7924 ± 0.0129
	'penalty': 'l2', 'C': 10	0.7026 ± 0.0093	0.7432 ± 0.0263	0.3544 ± 0.0130	0.2327 ± 0.0095	0.6976 ± 0.0104	0.7899 ± 0.0126
	'penalty': 'l2', 'C': 100	0.7020 ± 0.0098	0.7389 ± 0.0249	0.3526 ± 0.0130	0.2316 ± 0.0096	0.6975 ± 0.0104	0.7887 ± 0.0124
SVM	'kernel': 'linear', 'C': 1	0.6947 ± 0.0115	0.7474 ± 0.0261	0.3497 ± 0.0126	0.2283 ± 0.0092	0.6881 ± 0.0131	0.7856 ± 0.0126
	'kernel': 'linear', 'C': 10	0.6946 ± 0.0122	0.7488 ± 0.0285	0.3501 ± 0.0143	0.2286 ± 0.0107	0.6879 ± 0.0139	0.7857 ± 0.0138
	'kernel': 'linear', 'C': 100	0.6937 ± 0.0102	0.7495 ± 0.0269	0.3496 ± 0.0129	0.2281 ± 0.0095	0.6869 ± 0.0117	0.7855 ± 0.0123
	'kernel': 'rbf', 'C': 1	0.7180 ± 0.0115	0.7853 ± 0.0247	0.3796 ± 0.0140	0.2504 ± 0.0110	0.7097 ± 0.0133	0.8226 ± 0.0124
	'kernel': 'rbf', 'C': 10	0.7554 ± 0.0100	**0.8215 ± 0.0208**	**0.4247 ± 0.0156**	**0.2865 ± 0.0128**	0.7473 ± 0.0109	**0.8688 ± 0.0109**
	'kernel': 'rbf', 'C': 100	**0.7564 ± 0.0100**	0.8154 ± 0.0238	0.4238 ± 0.0142	0.2864 ± 0.0117	**0.7491 ± 0.0116**	0.8685 ± 0.0115
Decision Tree	**'criterion': 'entropy', 'max_depth': None**	**0.6990 ± 0.0126**	0.7899 ± 0.0243	**0.3658 ± 0.0152**	**0.2381 ± 0.0115**	**0.6878 ± 0.0141**	**0.7388 ± 0.0135**
	'criterion': 'entropy', 'max_depth': 5	0.6105 ± 0.0548	0.7234 ± 0.0699	0.2910 ± 0.0167	0.1832 ± 0.0158	0.5967 ± 0.0690	0.7070 ± 0.0135
	'criterion': 'entropy', 'max_depth': 10	0.6340 ± 0.0319	0.7750 ± 0.0431	0.3180 ± 0.0132	0.2004 ± 0.0115	0.6166 ± 0.0403	0.7341 ± 0.0117
	'criterion': 'gini', 'max_depth': None	0.6947 ± 0.0121	**0.7919 ± 0.0266**	0.3631 ± 0.0148	0.2356 ± 0.0111	0.6828 ± 0.0136	0.7373 ± 0.0142
	'criterion': 'gini', 'max_depth': 5	0.6112 ± 0.0514	0.7208 ± 0.0698	0.2904 ± 0.0148	0.1827 ± 0.0137	0.5977 ± 0.0654	0.7054 ± 0.0154
	'criterion': 'gini', 'max_depth': 10	0.6421 ± 0.0238	0.7800 ± 0.0408	0.3239 ± 0.0132	0.2047 ± 0.0103	0.6250 ± 0.0302	0.7271 ± 0.0148
Random Forest	'n_estimators': 50, 'max_depth': None	0.7980 ± 0.0109	0.8112 ± 0.0240	0.4689 ± 0.0189	0.3299 ± 0.0168	0.7964 ± 0.0122	0.8948 ± 0.0107
	'n_estimators': 50, 'max_depth': 5	0.6539 ± 0.0209	0.7697 ± 0.0255	0.3285 ± 0.0133	0.2090 ± 0.0108	0.6396 ± 0.0251	0.7785 ± 0.0119
	'n_estimators': 100, 'max_depth': None	0.7981 ± 0.0107	0.8149 ± 0.0246	0.4701 ± 0.0176	0.3305 ± 0.0157	0.7960 ± 0.0125	0.8980 ± 0.0107
	'n_estimators': 100, 'max_depth': 5	0.6539 ± 0.0184	0.7742 ± 0.0292	0.3297 ± 0.0149	0.2096 ± 0.0112	0.6390 ± 0.0216	0.7835 ± 0.0144
	'n_estimators': 200, 'max_depth': None	**0.8000 ± 0.0089**	**0.8214 ± 0.0245**	**0.4743 ± 0.0165**	**0.3336 ± 0.0145**	**0.7974 ± 0.0102**	**0.9013 ± 0.0110**
	'n_estimators': 200, 'max_depth': 5	0.6541 ± 0.0153	0.7732 ± 0.0249	0.3294 ± 0.0124	0.2094 ± 0.0092	0.6394 ± 0.0177	0.7822 ± 0.0138
GB	'n_estimators': 100, 'learning_rate': 0.1	0.7106 ± 0.0113	0.7760 ± 0.0214	0.3708 ± 0.0126	0.2437 ± 0.0099	0.7026 ± 0.0131	0.8131 ± 0.0118
	'n_estimators': 100, 'learning_rate': 0.01	0.6326 ± 0.0205	0.7319 ± 0.0380	0.3044 ± 0.0111	0.1924 ± 0.0083	0.6203 ± 0.0261	0.7419 ± 0.0137
	'n_estimators': 200, 'learning_rate': 0.1	0.7266 ± 0.0103	0.7897 ± 0.0200	0.3882 ± 0.0130	0.2575 ± 0.0106	0.7188 ± 0.0116	0.8278 ± 0.0100
	'n_estimators': 200, 'learning_rate': 0.01	0.6623 ± 0.0135	0.7426 ± 0.0326	0.3257 ± 0.0129	0.2087 ± 0.0093	0.6524 ± 0.0168	0.7664 ± 0.0139
	'n_estimators': 300, 'learning_rate': 0.1	**0.7361 ± 0.0101**	**0.8022 ± 0.0211**	**0.4004 ± 0.0133**	**0.2669 ± 0.0106**	**0.7279 ± 0.0113**	**0.8369 ± 0.0115**
	'n_estimators': 300, 'learning_rate': 0.01	0.6754 ± 0.0138	0.7492 ± 0.0276	0.3365 ± 0.0136	0.2171 ± 0.0104	0.6663 ± 0.0163	0.7800 ± 0.0123
XGBoost	'n_estimators': 100, 'learning_rate': 0.1	0.7510 ± 0.0120	0.8271 ± 0.0215	0.4220 ± 0.0156	0.2834 ± 0.0131	0.7416 ± 0.0135	0.8575 ± 0.0108
	'n_estimators': 100, 'learning_rate': 0.01	0.6631 ± 0.0210	0.7795 ± 0.0292	0.3373 ± 0.0154	0.2154 ± 0.0121	0.6488 ± 0.0249	0.7847 ± 0.0137
	'n_estimators': 200, 'learning_rate': 0.1	0.7668 ± 0.0093	0.8269 ± 0.0198	0.4379 ± 0.0143	0.2980 ± 0.0123	0.7594 ± 0.0105	0.8684 ± 0.0103
	'n_estimators': 200, 'learning_rate': 0.01	0.6901 ± 0.0141	0.7974 ± 0.0281	0.3612 ± 0.0133	0.2336 ± 0.0103	0.6769 ± 0.0171	0.8112 ± 0.0130
	'n_estimators': 300, 'learning_rate': 0.1	**0.7710 ± 0.0103**	**0.8297 ± 0.0213**	**0.4432 ± 0.0150**	**0.3025 ± 0.0130**	**0.7637 ± 0.0119**	**0.8749 ± 0.0108**
	'n_estimators': 300, 'learning_rate': 0.01	0.7084 ± 0.0137	0.8064 ± 0.0252	0.3780 ± 0.0147	0.2470 ± 0.0118	0.6963 ± 0.0159	0.8245 ± 0.0113
MLP	**'hidden_layer_sizes': (100,), 'activation': 'tanh'**	**0.7603 ± 0.0148**	**0.8303 ± 0.0214**	**0.4325 ± 0.0157**	**0.2926 ± 0.0141**	0.7517 ± 0.0177	**0.8612 ± 0.0103**
	'hidden_layer_sizes': (100,), 'activation': 'relu'	0.7598 ± 0.0133	0.8211 ± 0.0233	0.4291 ± 0.0162	0.2906 ± 0.0141	**0.7522 ± 0.0153**	0.8611 ± 0.0112
	'hidden_layer_sizes': (100,), 'activation': 'logistic'	0.7109 ± 0.0241	0.7470 ± 0.0353	0.3627 ± 0.0173	0.2399 ± 0.0154	0.7065 ± 0.0295	0.7990 ± 0.0125
	'hidden_layer_sizes': (100, 50, 25), 'activation': 'tanh'	0.7508 ± 0.0120	0.8215 ± 0.0252	0.4201 ± 0.0151	0.2823 ± 0.0126	0.7420 ± 0.0141	0.8467 ± 0.0121
	'hidden_layer_sizes': (100, 50, 25), 'activation': 'relu'	0.7521 ± 0.0108	0.8127 ± 0.0261	0.4187 ± 0.0151	0.2821 ± 0.0122	0.7446 ± 0.0126	0.8466 ± 0.0119
	'hidden_layer_sizes': (100, 50, 25), 'activation': 'logistic'	0.7278 ± 0.0336	0.7892 ± 0.0474	0.3908 ± 0.0292	0.2606 ± 0.0256	0.7203 ± 0.0402	0.8093 ± 0.0179
LightGBM	'n_estimators': 50, 'learning_rate': 0.1	0.7441 ± 0.0115	0.8048 ± 0.0244	0.4086 ± 0.0159	0.2740 ± 0.0129	0.7366 ± 0.0130	0.8441 ± 0.0109
	'n_estimators': 50, 'learning_rate': 0.01	0.6730 ± 0.0148	0.7388 ± 0.0323	0.3317 ± 0.0122	0.2140 ± 0.0092	0.6649 ± 0.0183	0.7656 ± 0.0155
	'n_estimators': 100, 'learning_rate': 0.1	0.7639 ± 0.0093	0.8299 ± 0.0216	0.4357 ± 0.0149	0.2956 ± 0.0124	0.7558 ± 0.0104	0.8657 ± 0.0107
	'n_estimators': 100, 'learning_rate': 0.01	0.6900 ± 0.0151	0.7526 ± 0.0305	0.3479 ± 0.0128	0.2264 ± 0.0100	0.6823 ± 0.0184	0.7863 ± 0.0135
	'n_estimators': 200, 'learning_rate': 0.1	**0.7731 ± 0.0092**	**0.8326 ± 0.0212**	**0.4463 ± 0.0147**	**0.3050 ± 0.0125**	**0.7657 ± 0.0104**	**0.8782 ± 0.0101**
	'n_estimators': 200, 'learning_rate': 0.01	0.7154 ± 0.0115	0.7691 ± 0.0253	0.3725 ± 0.0131	0.2459 ± 0.0103	0.7088 ± 0.0135	0.8115 ± 0.0117
(Meta-Model: LR)*	'C': 0.1, 'solver':liblinear, 'penalty': 'l1'	0.8201 ± 0.0104	0.8035 ± 0.0262	0.4954 ± 0.0202	0.3584 ± 0.0190	0.8222 ± 0.0119	0.9026 ± 0.0110
	'C': 0.1, 'solver':liblinear, 'penalty': 'l2'	0.8002 ± 0.0097	**0.8196 ± 0.0215**	0.4741 ± 0.0172	0.3337 ± 0.0154	0.7978 ± 0.0108	0.8997 ± 0.0102
	'C': 0.1, 'solver':saga, 'penalty': 'l1'	0.8240 ± 0.0103	0.7974 ± 0.0247	0.4990 ± 0.0181	0.3634 ± 0.0179	0.8273 ± 0.0120	0.9015 ± 0.0103
	'C': 0.1, 'solver':saga, 'penalty': 'l2'	0.8082 ± 0.0100	0.8123 ± 0.0210	0.4821 ± 0.0171	0.3430 ± 0.0161	0.8077 ± 0.0113	0.8987 ± 0.0096
	'C': 1, 'solver':liblinear, 'penalty': 'l1'	0.8257 ± 0.0102	0.7982 ± 0.0257	0.5016 ± 0.0181	0.3662 ± 0.0184	0.8291 ± 0.0124	0.9033 ± 0.0105
	'C': 1, 'solver':liblinear, 'penalty': 'l2'	0.8201 ± 0.0104	0.8075 ± 0.0218	0.4967 ± 0.0185	0.3590 ± 0.0179	0.8217 ± 0.0120	0.9037 ± 0.0104
	'C': 1, 'solver':saga, 'penalty': 'l1'	0.8265 ± 0.0096	0.7997 ± 0.0231	0.5033 ± 0.0178	0.3675 ± 0.0175	0.8299 ± 0.0111	0.9030 ± 0.0097
	'C': 1, 'solver':saga, 'penalty': 'l2'	0.8227 ± 0.0119	0.8043 ± 0.0223	0.4995 ± 0.0194	0.3626 ± 0.0198	0.8250 ± 0.0140	**0.9041 ± 0.0097**
	'C': 10, 'solver':liblinear, 'penalty': 'l1'	0.8262 ± 0.0087	0.7995 ± 0.0278	0.5027 ± 0.0174	0.3669 ± 0.0168	0.8296 ± 0.0106	0.9030 ± 0.0106
	'C': 10, 'solver':liblinear, 'penalty': 'l2'	0.8248 ± 0.0091	0.7968 ± 0.0262	0.4999 ± 0.0187	0.3644 ± 0.0171	0.8282 ± 0.0105	0.9014 ± 0.0109
	'C': 10, 'solver':saga, 'penalty': 'l1'	0.8255 ± 0.0092	0.7981 ± 0.0243	0.5013 ± 0.0154	0.3658 ± 0.0156	0.8289 ± 0.0115	0.9014 ± 0.0097
	'C': 10, 'solver':saga, 'penalty': 'l2'	0.8264 ± 0.0111	0.7959 ± 0.0214	0.5020 ± 0.0207	0.3670 ± 0.0202	0.8301 ± 0.0122	0.9006 ± 0.0108
	'C': 100, 'solver':liblinear, 'penalty': 'l1'	**0.8271 ± 0.0115**	0.7990 ± 0.0218	**0.5040 ± 0.0194**	**0.3685 ± 0.0200**	**0.8305 ± 0.0135**	0.9017 ± 0.0093
	'C': 100, 'solver':liblinear, 'penalty': 'l2'	0.8235 ± 0.0117	0.7995 ± 0.0227	0.4991 ± 0.0207	0.3631 ± 0.0205	0.8265 ± 0.0133	0.9016 ± 0.0109
	'C': 100, 'solver':saga, 'penalty': 'l1'	0.8247 ± 0.0107	0.7974 ± 0.0282	0.5000 ± 0.0202	0.3646 ± 0.0198	0.8281 ± 0.0126	0.9008 ± 0.0107
	'C': 100, 'solver':saga, 'penalty': 'l2'	0.8259 ± 0.0096	0.7983 ± 0.0275	0.5019 ± 0.0180	0.3663 ± 0.0169	0.8293 ± 0.0114	0.9022 ± 0.0114

*Note: The meta-model presented in this table is constructed by integrating the most effective models, each optimized with their respective best hyperparameters.

and $C = 10$, emerges as the best performer after analysis in every parameter obtaining the highest AUC (0.9717). The degree of capability of the model to differentiate levels of metric performance further demonstrates the confidence that would be placed on the particular model for such forecasting.

Random Forest and **GB**. The Random Forest with 100 estimators and GB with 300 estimators with no restriction on depth managed to give results. More worth noting, GB with *n_estimators* = *300* and *learning_rate* = *0.1* manages to strike a reasonable balance of ACC along with AUC, which is on the upper side at 96.70 % to be disregarded as excellent attributes to the model.

XGBoost predicting ability of the model, especially 300 estimators and a learning rate of 0.1, suggests an advantageous handle of the predictive power with an AUC reaching 0.9709. That efficiency in both PR and RC of the trend further indicates the user efficiency in targeting learners at different levels of academic performance.

Meta-Model. A LR-based meta-model where $C = 100$, *solver* = *'saga'*, *penalty* = *'l1'* outperformed all the individual

TABLE 9. The results of eight ML models are performed in various hyperparameters, particularly the Low-medium performance levels for Mathematics. In this case, modeled as an asterisk, this meta-model is built on all the optimized models implemented in this study, using an ensemble method to benefit from the best aspects of each model. Furthermore, the hypothetical and standard results of all examined measures are bootstraped with 100 repetitions, which is a solid practice in assessing how well the model performs.

Model	Hyperparameters	ACC	RC	F1S	PR	SP	AUC
LR **	'penalty': 'l1', 'C': 1	0.7594 ± 0.0058	0.7980 ± 0.0076	0.7663 ± 0.0054	**0.7370 ± 0.0064**	**0.7217 ± 0.0091**	**0.8394 ± 0.0050**
	'penalty': 'l1', 'C': 10	0.7578 ± 0.0056	0.7976 ± 0.0087	0.7650 ± 0.0058	0.7350 ± 0.0065	0.7188 ± 0.0083	0.8384 ± 0.0049
	'penalty': 'l1', 'C': 100	0.7585 ± 0.0059	0.7984 ± 0.0087	0.7657 ± 0.0061	0.7355 ± 0.0061	0.7194 ± 0.0080	0.8387 ± 0.0048
	'penalty': 'l2', 'C': 1	0.7588 ± 0.0054	0.7977 ± 0.0081	0.7658 ± 0.0056	0.7363 ± 0.0060	0.7208 ± 0.0075	0.8388 ± 0.0053
	'penalty': 'l2', 'C': 10	**0.7597 ± 0.0064**	**0.7993 ± 0.0086**	**0.7668 ± 0.0063**	0.7369 ± 0.0074	0.7209 ± 0.0100	0.8391 ± 0.0052
	'penalty': 'l2', 'C': 100	0.7579 ± 0.0057	0.7960 ± 0.0092	0.7647 ± 0.0056	0.7359 ± 0.0068	0.7206 ± 0.0097	0.8382 ± 0.0052
SVM	'kernel': 'linear', 'C': 1	0.7612 ± 0.0092	0.8164 ± 0.0072	0.7721 ± 0.0062	0.7323 ± 0.0084	0.7070 ± 0.0144	0.5160 ± 0.0082
	'kernel': 'linear', 'C': 10	0.7581 ± 0.0073	0.8241 ± 0.0093	0.7714 ± 0.0132	0.7251 ± 0.0136	0.6934 ± 0.0051	0.5013 ± 0.0101
	'kernel': 'linear', 'C': 100	0.7581 ± 0.0139	0.8146 ± 0.0104	0.7694 ± 0.0131	0.7290 ± 0.0140	0.7027 ± 0.0082	0.5027 ± 0.0061
	'kernel': 'rbf', 'C': 1	0.7907 ± 0.0055	0.8376 ± 0.0078	0.7986 ± 0.0141	0.7631 ± 0.0074	0.7447 ± 0.0064	0.5041 ± 0.0099
	'kernel': 'rbf', 'C': 10	0.8545 ± 0.0100	**0.8775 ± 0.0080**	0.8566 ± 0.0078	0.8368 ± 0.0054	0.8320 ± 0.0111	0.5182 ± 0.0100
	'kernel': 'rbf', 'C': 100	**0.8681 ± 0.0102**	0.8734 ± 0.0120	**0.8677 ± 0.0086**	**0.8621 ± 0.0147**	**0.8628 ± 0.0146**	**0.5196 ± 0.0075**
Decision Tree	**'criterion': 'entropy', 'max_depth': None**	**0.8382 ± 0.0061**	**0.8419 ± 0.0084**	**0.8372 ± 0.0065**	0.8327 ± 0.0066	0.8346 ± 0.0068	**0.8383 ± 0.0062**
	'criterion': 'entropy', 'max_depth': 5	0.6982 ± 0.0070	0.7504 ± 0.0478	0.7103 ± 0.0135	0.6765 ± 0.0161	0.6472 ± 0.0485	0.7665 ± 0.0063
	'criterion': 'entropy', 'max_depth': 10	0.7311 ± 0.0070	0.8169 ± 0.0228	0.7501 ± 0.0081	0.6939 ± 0.0105	0.6473 ± 0.0242	0.8004 ± 0.0064
	'criterion': 'gini', 'max_depth': None	0.8374 ± 0.0054	0.8396 ± 0.0072	0.8362 ± 0.0055	**0.8329 ± 0.0072**	**0.8353 ± 0.0078**	0.8374 ± 0.0054
	'criterion': 'gini', 'max_depth': 5	0.6992 ± 0.0070	0.7560 ± 0.0444	0.7126 ± 0.0133	0.6758 ± 0.0152	0.6438 ± 0.0442	0.7659 ± 0.0066
	'criterion': 'gini', 'max_depth': 10	0.7371 ± 0.0067	0.8250 ± 0.0231	0.7561 ± 0.0079	0.6983 ± 0.0105	0.6511 ± 0.0237	0.8001 ± 0.0059
Random Forest	'n_estimators': 50, 'max_depth': None	0.8784 ± 0.0044	0.8983 ± 0.0062	0.8796 ± 0.0044	**0.8617 ± 0.0068**	**0.8590 ± 0.0076**	0.9548 ± 0.0022
	'n_estimators': 50, 'max_depth': 5	0.7173 ± 0.0059	0.8833 ± 0.0132	0.7554 ± 0.0053	0.6600 ± 0.0069	0.5551 ± 0.0167	0.8084 ± 0.0065
	'n_estimators': 100, 'max_depth': None	0.8814 ± 0.0044	0.9066 ± 0.0054	0.8832 ± 0.0043	0.8610 ± 0.0067	0.8569 ± 0.0078	0.9575 ± 0.0023
	'n_estimators': 100, 'max_depth': 5	0.7189 ± 0.0063	0.8820 ± 0.0107	0.7562 ± 0.0051	0.6619 ± 0.0070	0.5594 ± 0.0154	0.8088 ± 0.0059
	'n_estimators': 200, 'max_depth': None	**0.8828 ± 0.0043**	**0.9097 ± 0.0066**	**0.8847 ± 0.0043**	0.8611 ± 0.0064	0.8565 ± 0.0075	**0.9587 ± 0.0022**
	'n_estimators': 200, 'max_depth': 5	0.7202 ± 0.0070	0.8842 ± 0.0088	0.7575 ± 0.0056	0.6626 ± 0.0075	0.5598 ± 0.0142	0.8102 ± 0.0054
GB	'n_estimators': 100, 'learning_rate': 0.1	0.7585 ± 0.0056	0.8156 ± 0.0084	0.7695 ± 0.0053	0.7284 ± 0.0064	0.7027 ± 0.0091	0.8395 ± 0.0052
	'n_estimators': 100, 'learning_rate': 0.01	0.7046 ± 0.0068	**0.8654 ± 0.0234**	0.7432 ± 0.0074	0.6517 ± 0.0087	0.5475 ± 0.0245	0.7876 ± 0.0070
	'n_estimators': 200, 'learning_rate': 0.1	0.7689 ± 0.0058	0.8153 ± 0.0087	0.7772 ± 0.0058	0.7424 ± 0.0067	0.7235 ± 0.0092	0.8503 ± 0.0056
	'n_estimators': 200, 'learning_rate': 0.01	0.7235 ± 0.0062	0.8554 ± 0.0118	0.7536 ± 0.0059	0.6736 ± 0.0065	0.5947 ± 0.0122	0.8035 ± 0.0066
	'n_estimators': 300, 'learning_rate': 0.1	**0.7746 ± 0.0054**	0.8162 ± 0.0082	**0.7816 ± 0.0054**	**0.7499 ± 0.0060**	**0.7339 ± 0.0078**	**0.8567 ± 0.0045**
	'n_estimators': 300, 'learning_rate': 0.01	0.7335 ± 0.0060	0.8376 ± 0.0090	0.7565 ± 0.0055	0.6898 ± 0.0066	0.6317 ± 0.0112	0.8135 ± 0.0054
XGBoost	'n_estimators': 100, 'learning_rate': 0.1	0.7949 ± 0.0056	0.8447 ± 0.0073	0.8028 ± 0.0055	0.7649 ± 0.0068	0.7462 ± 0.0084	0.8758 ± 0.0051
	'n_estimators': 100, 'learning_rate': 0.01	0.7384 ± 0.0066	0.8356 ± 0.0122	0.7594 ± 0.0060	0.6961 ± 0.0082	0.6433 ± 0.0150	0.8178 ± 0.0064
	'n_estimators': 200, 'learning_rate': 0.1	0.8176 ± 0.0052	0.8581 ± 0.0074	0.8230 ± 0.0051	0.7907 ± 0.0064	0.7779 ± 0.0081	0.8950 ± 0.0043
	'n_estimators': 200, 'learning_rate': 0.01	0.7509 ± 0.0054	0.8449 ± 0.0088	0.7702 ± 0.0052	0.7078 ± 0.0063	0.6590 ± 0.0093	0.8321 ± 0.0052
	'n_estimators': 300, 'learning_rate': 0.1	**0.8329 ± 0.0057**	**0.8690 ± 0.0075**	**0.8372 ± 0.0055**	**0.8076 ± 0.0072**	**0.7976 ± 0.0089**	**0.9068 ± 0.0042**
	'n_estimators': 300, 'learning_rate': 0.01	0.7619 ± 0.0063	0.8428 ± 0.0098	0.7777 ± 0.0063	0.7221 ± 0.0068	0.6829 ± 0.0096	0.8436 ± 0.0055
MLP	**'hidden_layer_sizes': (100,), 'activation': 'tanh'**	**0.8615 ± 0.0057**	**0.8657 ± 0.0101**	**0.8607 ± 0.0061**	**0.8559 ± 0.0076**	**0.8574 ± 0.0085**	**0.9139 ± 0.0046**
	'hidden_layer_sizes': (100,), 'activation': 'relu'	0.8561 ± 0.0074	0.8577 ± 0.0112	0.8549 ± 0.0067	0.8525 ± 0.0129	0.8546 ± 0.0178	0.8985 ± 0.0046
	'hidden_layer_sizes': (100,), 'activation': 'logistic'	0.8490 ± 0.0062	0.8569 ± 0.0149	0.8486 ± 0.0069	0.8408 ± 0.0106	0.8412 ± 0.0137	0.9010 ± 0.0048
	'hidden_layer_sizes': (100, 50, 25), 'activation': 'tanh'	0.8571 ± 0.0048	0.8586 ± 0.0077	0.8559 ± 0.0052	0.8532 ± 0.0061	0.8556 ± 0.0070	0.9017 ± 0.0047
	'hidden_layer_sizes': (100, 50, 25), 'activation': 'relu'	0.8565 ± 0.0051	0.8586 ± 0.0072	0.8554 ± 0.0054	0.8523 ± 0.0066	0.8545 ± 0.0072	0.8971 ± 0.0047
	'hidden_layer_sizes': (100, 50, 25), 'activation': 'logistic'	0.8525 ± 0.0056	0.8542 ± 0.0094	0.8513 ± 0.0059	0.8485 ± 0.0087	0.8508 ± 0.0102	0.8857 ± 0.0051
LightGBM	'n_estimators': 50, 'learning_rate': 0.1	0.7731 ± 0.0058	0.8187 ± 0.0083	0.7810 ± 0.0059	0.7467 ± 0.0069	0.7285 ± 0.0091	0.8544 ± 0.0050
	'n_estimators': 50, 'learning_rate': 0.01	0.7292 ± 0.0062	0.8032 ± 0.0151	0.7456 ± 0.0069	0.6960 ± 0.0073	0.6569 ± 0.0142	0.8034 ± 0.0054
	'n_estimators': 100, 'learning_rate': 0.1	0.7923 ± 0.0062	0.8299 ± 0.0078	0.7980 ± 0.0061	0.7685 ± 0.0074	0.7556 ± 0.0091	0.8734 ± 0.0049
	'n_estimators': 100, 'learning_rate': 0.01	0.7417 ± 0.0063	0.8192 ± 0.0114	0.7582 ± 0.0065	0.7057 ± 0.0070	0.6660 ± 0.0112	0.8181 ± 0.0058
	'n_estimators': 200, 'learning_rate': 0.1	**0.8175 ± 0.0050**	**0.8521 ± 0.0074**	**0.8220 ± 0.0050**	**0.7939 ± 0.0060**	**0.7838 ± 0.0074**	**0.8946 ± 0.0042**
	'n_estimators': 200, 'learning_rate': 0.01	0.7521 ± 0.0049	0.8253 ± 0.0083	0.7669 ± 0.0048	0.7164 ± 0.0062	0.6806 ± 0.0092	0.8307 ± 0.0044
(Meta-Model: LR)*	'C': 0.1, 'solver':liblinear, 'penalty': 'l1'	0.8816 ± 0.0046	0.8980 ± 0.0074	0.8827 ± 0.0044	0.8679 ± 0.0054	0.8655 ± 0.0066	0.9594 ± 0.0027
	'C': 0.1, 'solver':liblinear, 'penalty': 'l2'	0.8841 ± 0.0046	0.8977 ± 0.0094	0.8848 ± 0.0049	0.8724 ± 0.0032	0.8709 ± 0.0036	0.9597 ± 0.0021
	'C': 0.1, 'solver':saga, 'penalty': 'l1'	0.8846 ± 0.0045	**0.9011 ± 0.0064**	0.8856 ± 0.0046	0.8706 ± 0.0048	0.8683 ± 0.0048	0.9610 ± 0.0026
	'C': 0.1, 'solver':saga, 'penalty': 'l2'	**0.8858 ± 0.0042**	0.8984 ± 0.0058	**0.8863 ± 0.0045**	**0.8747 ± 0.0052**	0.8734 ± 0.0057	0.9608 ± 0.0012
	'C': 1, 'solver':liblinear, 'penalty': 'l1'	0.8839 ± 0.0032	0.8963 ± 0.0040	0.8845 ± 0.0032	0.8730 ± 0.0045	0.8717 ± 0.0050	0.9604 ± 0.0017
	'C': 1, 'solver':liblinear, 'penalty': 'l2'	0.8844 ± 0.0034	0.8988 ± 0.0050	0.8852 ± 0.0034	0.8720 ± 0.0062	0.8702 ± 0.0071	0.9612 ± 0.0013
	'C': 1, 'solver':saga, 'penalty': 'l1'	0.8850 ± 0.0047	0.8997 ± 0.0047	0.8858 ± 0.0046	0.8724 ± 0.0067	0.8706 ± 0.0072	0.9611 ± 0.0020
	'C': 1, 'solver':saga, 'penalty': 'l2'	0.8855 ± 0.0030	0.8987 ± 0.0068	0.8862 ± 0.0035	0.8740 ± 0.0040	0.8725 ± 0.0047	**0.9613 ± 0.0024**
	'C': 10, 'solver':liblinear, 'penalty': 'l1'	0.8843 ± 0.0038	0.8950 ± 0.0049	0.8846 ± 0.0039	0.8745 ± 0.0043	**0.8737 ± 0.0044**	0.9610 ± 0.0026
	'C': 10, 'solver':liblinear, 'penalty': 'l2'	0.8826 ± 0.0037	0.8978 ± 0.0049	0.8835 ± 0.0038	0.8697 ± 0.0059	0.8677 ± 0.0061	0.9606 ± 0.0018
	'C': 10, 'solver':saga, 'penalty': 'l1'	0.8846 ± 0.0052	0.9007 ± 0.0072	0.8856 ± 0.0052	0.8711 ± 0.0057	0.8689 ± 0.0061	0.9611 ± 0.0028
	'C': 10, 'solver':saga, 'penalty': 'l2'	0.8829 ± 0.0035	0.8938 ± 0.0058	0.8833 ± 0.0038	0.8730 ± 0.0048	0.8721 ± 0.0048	0.9603 ± 0.0023
	'C': 100, 'solver':liblinear, 'penalty': 'l1'	0.8804 ± 0.0047	0.8943 ± 0.0087	0.8812 ± 0.0049	0.8685 ± 0.0074	0.8667 ± 0.0089	0.9591 ± 0.0024
	'C': 100, 'solver':liblinear, 'penalty': 'l2'	0.8838 ± 0.0036	0.8980 ± 0.0073	0.8846 ± 0.0038	0.8717 ± 0.0048	0.8699 ± 0.0055	0.9601 ± 0.0024
	'C': 100, 'solver':saga, 'penalty': 'l1'	0.8818 ± 0.0026	0.8954 ± 0.0058	0.8825 ± 0.0028	0.8700 ± 0.0050	0.8684 ± 0.0056	0.9596 ± 0.0017
	'C': 100, 'solver':saga, 'penalty': 'l2'	0.8818 ± 0.0036	0.8956 ± 0.0042	0.8825 ± 0.0034	0.8699 ± 0.0045	0.8682 ± 0.0052	0.9600 ± 0.0015

* Note: The meta-model presented in this table is constructed by integrating the most effective models, each optimized with their respective best hyperparameters.
** Note: The LR model with *penalty= 'l2'*, *C= 10* demonstrates superior performance, underlined by leading results in three critical metrics (ACC, RC, and F1S) and bolded as the best option in terms of PR, SP, and AUC. This configuration stands out not only for its slightly higher metric averages but also for its underlying results, indicating robustness across different evaluation criteria. Despite the close competition with *penalty = 'l1'*, *C= 1*, the nuanced advantage in critical areas (highlighted by the underline and bold emphasis) establishes it as the optimal choice in scenarios prioritizing a balanced trade-off between ACC and RC, alongside exceptional specificity and AUC performance.

models with the highest AUC of 0.9775. This underscores the ability of the metamodel to leverage the strengths of the individual expert models to achieve more accurate predictions.

C. MEDIUM AND HIGH-PERFORMANCE STUDY

The results displayed in Table 8 further point out a variation in the performance across several metrics of interest, such as ACC, RC, F1S, PR, SP, and AUC, stemming from the models abilities to tell the difference between Medium and High achievers. This investigation covers the range of hyperparameter configurations for each model and a new combined model, the so-called meta-model. The meta-model *, which combines the informative capacity of the stands' models, is composed of the best possible models for this particular

TABLE 10. Comparative analysis of ML models using confusion matrices across three performance scenarios: Low-Medium, High-Medium, and Low-High student achievement levels.

Each cell: Confusion Matrix (rows: True Class — True Negative, True Positive; columns: Predicted Class — True Negative, True Positive).

Model	Low-High	Medium-High	Low-Medium
LR	2464 326 / 34 303	1911 818 / 81 256	2032 758 / 523 2206
SVM	2420 370 / 32 305	1824 905 / 103 234	1898 892 / 760 1969
Decision Tree	2146 644 / 80 257	1636 1093 / 122 215	1780 1010 / 920 1809
Random Forest	2389 401 / 31 306	1871 858 / 94 243	1939 851 / 528 2201
GB	2429 361 / 29 308	1869 860 / 90 247	2020 770 / 509 2220
XGBoost	2423 367 / 36 301	1863 866 / 95 242	2022 768 / 557 2172
MLP	2489 301 / 43 294	1881 848 / 115 222	1897 893 / 826 1903
LightGBM	2425 365 / 31 306	1886 843 / 102 235	2037 753 / 565 2164
Meta-Model: LR	2425 365 / 31 306	1918 811 / 82 255	2066 724 / 550 2179

analysis, i.e., perfectly fine-tuned to the hyperparameters of each model. Coalescing all the distinct characteristics of each stand model with optimal hyperparameters further improves the model predictive power. Several essential remarks on the composed tables result from the conclusions made in the previous section:

SVM with RBF Kernel. This model, especially with $C = 10$, performs the best among the rest of the models with dozens of metrics. In particular, it has the best AUC among the other models. This evidence leads to the conclusion about the discriminative ability of the SVM over the two performance levels, which in turn shows why it can be employed as the best model for this prediction task.

Random Forest using 200 estimators and no limit on max depth is quite optimal and very close to the SVM model. This model is notable for its reasonable balance between PR and RC, which means it is efficient at catching High achievers while keeping false alarms to a minimum.

GB and **XGBoost**. Most appeared efficient in the paperwork, especially with higher values of *n_estimators* = *300* and *learning_rate* = *0.1*. XGboost is a bit more productive, especially in AUC measures while measuring efficiency in prediction rank, than XGBoost, which uses their corresponding learning rate 251.

Meta-Model Performance. LR-based meta-model (C = *100*, *penalty* = *'l1'*, *solver* = *'liblinear'*) can beat several individual models and many other metrics, including F1S and AUC. This highlights the necessity of the average as mentioned above because some generally accepted model reconstructing aiming solely for predicting should not exist.

Considerations on Model Selection. Although the meta-model performs markedly better in terms of some metrics, the metric alone cannot be a determinant in selecting the best model. The decision should incorporate the nature of the problem where the model will be used, high tolerance for an FP (PR), low threshold of missing high performers (RC), and overall ACC. The performance of the meta-model appeals to the balanced approach taken on the metric, making the model desirable in situations where the trade-off between PR and RC is needed.

D. LOW AND MEDIUM PERFORMANCE STUDY

The assessment detailed in Table 9 is well and truly a thorough measurement of model performance across parameters such as ACC, RC, F1S, PR, SP, and AUC. In addition, there is also an engagement of the novel approach but indicated with an asterisk * whereby all the efforts in the individual models are optimized to the best parameters only for this multi-level study. The meta-model approach further improves ACC and generalization performance by combining the best characteristics of all the models. While the tabulated results have been analyzed, some critical issues of the table are brought out:

1) LR

The best performance was noted with the 'l2' penalty and C = *10*, in which case a sound measure of regularization should be applied to not fit the model too much and still get an effective measure of accuracy. Such results indicate LR efficacy when properly set for these complicated tasks.

2) SVM

In the SVM model, the performance is not as optimal as projected, even with the RBF kernel and C = *100*. These results mean that SVMs are potent tools but may require proper fine-tuning of parameters and may not be suitable for all Low to Medium levels.

3) DECISION TREES

Performance was optimal when the depth of the model was unconstrained, and the 'entropy' criterion was utilized, which points towards the model attention to interplay with various complex facets of the given data when the model is allowed to branch out entirely. Such findings bring out the prospect of decision trees in such cases of the application where the need to understand complex relationships within the data is more important.

4) RANDOM FOREST

In this case, however, striking performances were achieved with the 200 estimators used against the estimators' no depth restriction aspect, underlining the contribution that aggregation of decisions has on the accuracy of the predictions. This performance demonstrates the value of Random Forests in reducing variance and preventing overfitting.

5) META-MODEL

In particular, the configuration of hyperparameters is as follows: C=*0.1*, *solver*=*'saga'*, and *penalty*=*'l2'*. This specific setup was selected as it has reported high ACC, RC, and F1S and strong PR, SP, and AUC values. The ensemble approach combines the strengths of all other models and raises the strength and predicting power across all evaluation metrics.

6) MODEL SELECTION AND HYPERPARAMETER OPTIMIZATION

This paper highlights that hyperparameter optimization should be undertaken with the aim of eliciting maximally favorable responses from each of the models. When determining which modeling approach best suits the particular case, attention should be paid to its requirements in aspects such as PR, RC, or any other accuracy measure necessary. The even performance of the meta-model across most and especially the relevant elements we care about makes it a model of choice for applications with many handle decisions.

E. CONFUSION MATRICES

As illustrated in Table 10, a comparative analysis of ML models has been conducted using confusion matrices across three performance scenarios: Low-High, Medium-High, and Low-Medium achievement levels. Results of the performance of several ML models are presented, among them LR, SVM, Decision Tree, Random Forest, GB, XGBoost, MLP, and LightGBM with an LR-based meta-model.

The quantitative performance of the models is represented through confusion matrix values, which include the number of TP, the number of TN, the number of FP, and the number of FN. Such metrics make it possible to rank the accuracy of the assumed anthropogenic influence. For instance, the LR model exhibits TP counts of 22666, 911, and 523 for Low-High, Medium-High, and Low-Medium scenarios, respectively. However, in the case of the corroborative factors scenario, the model XGBoost performs better, with a slight improvement in the TP counts for the same scenarios. The differences among the various models point to the differences in effectiveness given the complex nature of educational data models.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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